Healthcare Privatization and Patient Health: Evidence from Puerto Rico*

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Abstract

Public health systems increasingly rely on private providers to deliver publicly financed care, yet the consequences of large-scale privatization for patient health remain poorly understood. This paper examines Puerto Rico's 1994-2000 health reform, which transformed a predominantly publicly operated health system into one relying on private providers, while simultaneously changing provider payment incentives and eligibility. I assemble a novel dataset combining employment, establishment, and wage records from the Quarterly Census of Employment and Wages; hospital data from Medicare provider files; and mortality and birth records from the Puerto Rico Planning Board and newly digitized vital-statistics reports. Exploiting the rollout of the reform across counties and using the staggered difference-in-differences estimator proposed by Callaway and Sant'Anna (2021), I find that the reform increased all-cause mortality by 3.7 percent relative to counties not yet reformed. The reform also reduced ambulatory-care employment, the number of hospitals, and the use of early prenatal care, suggesting that declining provider capacity and reduced access to timely care may have contributed to worsening health outcomes. These findings highlight the potential health costs of large-scale transitions to privately delivered care and underscore the importance of maintaining provider capacity and access when restructuring publicly financed health systems.

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1 Introduction

Health spending has risen steadily as a share of GDP across OECD countries, averaging 9.3 percent in 2024 (roughly double its 1970 level), and reaching 17.2 percent in the United States (Organisation for Economic Co-operation and Development, 2025). Governments finance most of this spending. Across the OECD, government and compulsory schemes cover close to three-quarters of health costs, and in the United States Medicare, Medicaid, and the Children's Health Insurance Program (CHIP) together accounted for about 7 percent of GDP in 2024 (Medicaid and CHIP Payment and Access Commission, 2026; Centers for Medicare and Medicaid Services, 2025a). To reduce costs, governments have increasingly relied on the private sector to manage their healthcare. There is a rich literature that examines these effects at small scale and for specific aspects of privatization (Fabre and Straub, 2023). I contribute to this literature by studying this shift at an unusually large scale, examining Puerto Rico's 1994 health reform, which transferred the management of its extensive public health care system to the private sector.

Before the reform, Puerto Rico ran a system of health care public provision, in which it (i) owned the facilities, (ii) employed salaried providers, and (iii) provided universal coverage (for free), since any resident could receive care in its facilities. The reform changed all three. It (i) sold or leased the facilities to private providers, (ii) replaced government provider salaries with private-sector capitated contracts, and (iii) replaced universal coverage with a government health insurance with co-payments for residents below 200 percent of the Federal poverty line, administered by private Medicaid managed-care organizations (MCOs) (Commonwealth of Puerto Rico, 1993). The reform sought to decrease government health care costs, an objective reflected in the decline in annual spending growth from approximately 6 percent in 1985–1993 to less than 1 percent during the rollout (Figure 1). Implementation proceeded in six cohorts, extending the reform across all 78 counties between 1994 and 2000.

Several aspects of the reform's implementation raised concerns about access to care and

patient health. By 2001, closures and reduced operating hours at privatized facilities had left only 58 of the island's 78 counties with round-the-clock emergency services (Pan American Health Organization, 2007). Approximately ten primary-care centers sold to private providers had also begun closing facilities or reducing service hours, prompting the government to subsidize their operations or reacquire facilities (Pan American Health Organization, 2007). In 2003, the legislature stated that the sales had left residents uncertain about where to seek care (**?**). The shift to capitated payments also raised concerns about incentives to restrict referrals and services. A 2005 evaluation warned that these incentives could lead physicians to withhold indicated care or delay clinically necessary tests. It also reported the Office of the Patient Advocate's assessment that the system rationed services to decrease its costs and that its predominantly acute-care orientation came at the expense of preventive care (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005; Pan American Health Organization, 2007).

The health care workforce changed as well. Positions in government health care facilities and regulatory agencies fell by 33 percent from 1996 to 1999 (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1998), and the number of licensed physicians declined by 9.7 percent between 1992 and 2001 (Appendix Table A1)[1]. When it comes to patients, mean county all-cause mortality increased by approximately 4.6%, from 716 deaths per 100,000 residents in the pre-reform period (1990–1993) to 749 during the reform rollout (1994–2000) (see Table 1). The share of household consumption devoted to medical care continued its same pre-reform upward trend (Appendix Figure A1), which I interpret as indicative that patient costs and healthcare-seeking behavior remained largely unchanged. The drastic changes in provider capacity and their payment incentives, along with changes in mortality rates, suggest that the reform affected patients' health primarily through the restructuring of health care delivery.

[1]Contextualized with the increase in the number of employees under the Ambulatory Health Care Services industry (which increased by 67.9% between 1992 and 2000), the decline in physicians may indicate an increase in administrative staff, now required to manage the expanding MCOs (see Table A2).

A previous study estimates the reform's effects on mental health care use, in a survey of about 3,000 adults (Alegría et al., 2001).[2] To expand the understanding of this health reform, I measure its effects on mortality for all residents of Puerto Rico, providing a direct measure of whether these privatization efforts ultimately affected patients' survival. I build a county-year panel for Puerto Rico's 78 counties from five sources. The Puerto Rico Planning Board provides annual population and deaths by cause (Junta de Planificación de Puerto Rico, 2026). The Department of Health's annual vital-statistics reports, digitized for this paper for 1990 to 2002, add births by hospital ownership, prenatal care, and birth weight (Puerto Rico Department of Health, 2025). The death records of Puerto Rico's Demographic Registry, of which the Puerto Rico Institute of Statistics publishes a randomly selected sample of 50% of registered deaths, add the age and sex of each person who died (Instituto de Estadísticas de Puerto Rico, 2026). The Quarterly Census of Employment and Wages measures ambulatory health care employment, establishments, and wages (U.S. Bureau of Labor Statistics, 2025), and the Provider of Services files of the Centers for Medicare and Medicaid Services identify Medicare-certified hospitals (Centers for Medicare and Medicaid Services, 2025b). Mortality for mainland counties in states within the United States that had not adopted Medicaid managed care by 2000, from the Centers for Disease Control and Prevention, provides a second comparison group for robustness (National Center for Health Statistics, 2026).

There are many advantages in the study design of this setting. First, because the reform was implemented in waves across Puerto Rico's 78 counties, I can compare counties that have already undergone the reform with those not yet treated, providing a natural comparison between treated and control counties. Additionally, I rely on the estimator of Callaway and Sant'Anna (2021) to account for heterogeneous treatment effects across cohorts and years, which proves to be robust and gives estimates similar to those of other heterogeneity-robust difference-in-differences estimators. Furthermore, MCOs were assigned to counties, and res-

[2]Their results are mixed, with no significant change in use among poor adults and a rise in specialty use among the non-poor.

idents that were beneficiaries of Medicaid had no choice of selection among insurers (since there was only one assigned per county), which decreases endogenous issues on self-selection. Because Puerto Rico is an island, no mainland provider is within reach for routine care, so cross-state-border patient flows are absent. Moreover, contrary to states in the United States, Puerto Rico's federal Medicaid funding is subject to an annual cap, leaving the local government responsible for additional spending once the federal allotment is exhausted (Mitchell, 2025). Thus, federal matching-rate increases or Medicaid expansions do not generate additional federal funds, limiting the potential for federal policy changes to confound the estimated effects of Puerto Rico's health reform. Additionally, since self-enrollment required applicants to verify their county of residence, this prevented beneficiaries of government health care that were residents of the not-yet-treated counties from seeking health care in treated counties. However, those from treated counties could still seek health care from not-yet-treated counties. Yet, U.S. Census commuting data indicates that approximately 70% of work commutes remain within the same reform cohort (Appendix Figure A2, panel (a)).

The reform most directly affected residents without private insurance, Medicare, or coverage through the public employees' insurance plan; these residents accounted for 51 percent of the population in 1990 and 47 percent in 2003 (Table 2). I estimate the cross-county spillover to be driven by a maximum of 4 percent[3] to 0% of workers living in treated counties (if none of them do)[4]. This small share of residents could bring a downwards bias to the estimated mortality rate, if treated residents avoided adverse changes in their health care by seeking public health care in not-yet-treated counties. This is because health care in treated cohorts was aimed at decreasing utilization, relative to the not-yet-treated cohort. The estimates would therefore be conservative.

Privatization has been associated with higher mortality (Picone et al., 2002; Knutsson and Tyrefors, 2022; Goodair and Reeves, 2022), with elderly patients faring worse (Duggan et al.,

[3]This assumes that every uncovered worker living in a treated county who worked in a not-yet-treated county sought public health care there.

[4]For estimation details, refer to Appendix E.1.

2023). Medicaid managed care has also been found to reduce preventive care utilization, specifically prenatal care, and raise low birth weight and neonatal death (Aizer et al., 2007). In Texas, the switch to Medicaid managed care raised mortality and preterm birth among children of Black mothers, a higher-cost group, and lowered both among children of U.S.-born Hispanic mothers (Kuziemko et al., 2018). Public providers have been found to match or outperform private ones in quality of care for dually eligible U.S. veterans (Chan et al., 2023) and after hospital conversions to private ownership (Goodair and Reeves, 2024). In other contexts, such as Germany, in-hospital mortality did not rise at hospitals converted to for-profit status (Tiemann and Schreyögg, 2012). Other studies link higher mortality to reductions in staffing ratios following for-profit hospital conversions (Picone et al., 2002), declines in ambulance staff quality (Knutsson and Tyrefors, 2022), and decreases in staffing per patient and Medicaid patient admissions following the privatization of U.S. government hospitals (Duggan et al., 2023).

I interpret this evidence and my estimates with an incomplete-contracts model of public and private management adapted from Hart et al. (1997) (Appendix C).[5] Because patient needs vary and many dimensions of quality are hard to measure and verify, contracts do not specify how care should be delivered in every circumstance. Whoever holds residual operational control makes the decisions that contracts leave open, and the health reform moved that control from the government to private managers. The model has four stages. First, the government and the manager sign a contract that fixes the manager's pay. Second, the manager chooses non-contractible effort toward two kinds of innovation. A cost cut saves money and can harm patients, and a quality improvement that raises the value of care to patients and increased costs. Third, the manager carries out its innovations. Fourth, it delivers care and collects its payoff.

[5]Coube et al. (2026) study Brazil's Organizações Sociais de Saúde (OSS) contracts, which give nonprofit managers control over staffing and procurement in public hospitals but require them to reinvest any budgetary savings. They read the arrangement as public ownership with private management in the sense of Hart et al. (1997). In the model's terms, the contract leaves the manager no private gain from cutting costs. They find higher admissions with no detectable change in risk-adjusted inpatient mortality.

A private manager is paid a fixed amount per enrollee and bears the cost of care, so it keeps whatever a cost cut saves and bargains with the government only over quality improvements (since it increases costs), receiving half of their net benefit. A public manager is paid a fixed salary while the government bears the cost of care, so the savings go to the government, and the manager bargains with it over both innovations, receiving half of whatever part of their value a replacement could not deliver. A manager is easy to replace when that part is small. Because the government represents patients, it values a cut at its savings minus the harm, so a cut that harms patients as much as it saves earns the public manager nothing and the private manager the full savings. Anticipating these payoffs at the second stage, the private manager cuts costs past the efficient level and makes more quality improvements than a public manager to the extent that public managers can be replaced. Whether patients gain or lose therefore depends on which of these two effects is larger.

I hypothesize that mortality is higher under private control when the harm from the additional cost-cutting exceeds the value to patients of the additional quality improvement, with each service weighted by its importance for survival. The condition holds whenever public managers are hard enough to replace. The model leaves the direction of staffing and facility changes open, because quality improvement can add inputs that cost-reductions removes. In this model, I hold cost sharing, covered benefits, and eligibility fixed.[6]

Results indicate that relative to counties the reform had not yet treated, all-cause mortality rose by 3.7 percent of its pre-treatment mean, consistent with the hypothesis. The supply estimates are consistent with the mechanisms the literature documents, where staffing and the capacity to deliver care decrease. Ambulatory health care employment was 9.6 percent below its counterfactual path, and the counties the reform treated had 6.5 percent fewer ambulatory establishments and 3.4 percent fewer hospitals per capita, relative to counties not yet treated. The share of mothers receiving early prenatal care also fell by 3.0 per-

[6]I hold eligibility fixed because the reform left a small share of residents uncovered (Appendix C), and increased cost sharing and benefits by small amounts.

cent. The increase falls on working-age adults, whose deaths rose by 11.2 percent of their pre-treatment mean; deaths of residents aged 65 and over, who qualified for Medicare, did not change significantly. The increase in all-cause mortality stays consistent under different robustness checks.

Research on health care privatization compares public and private providers that operate within the same geographic area or examines outsourcing within mixed systems (Chan et al., 2023; Goodair and Reeves, 2022), as well as ownership changes at some hospitals (Duggan et al., 2023; Tiemann and Schreyögg, 2012). This paper complements this literature by examining a transition to private management across a large public delivery system. Patients may turn to other providers when one hospital privatizes, but a system-wide transfer leaves fewer public providers to turn to. In this sense spillovers are minimized in my setting. The reform also speaks to research on payment incentives and insurance coverage. Its transition from salaried public provision to capitated private management extends evidence on Medicaid managed care drawn from comparisons with fee-for-service arrangements (Duggan, 2004; Duggan et al., 2018; Aizer et al., 2007). Additionally, its replacement of universal eligibility for public services with income-based insurance eligibility also relates to research on the health effects of health care coverage (Finkelstein et al., 2012). With a reported 93 percent of residents insured by 2003 (3 years after the reform was implemented), this setting allows me to examine the health consequences of restructuring care in a largely insured population. The estimates capture the combined effects of changes in ownership, provider payment incentives, and eligibility requirements.

To my knowledge, this paper provides the first causal estimates of the reform's effects on patient health. Results are informative for governments shifting public health care from public to private providers.

2 Setting

2.1 Before the reform

As of 1956, Puerto Rico managed its public health care program through the integrated, regionalized system described by Arbona and Ramírez de Arellano (1978). Government facilities and salaried staff provided local care through 78 Diagnostic and Treatment Centers (CDTs), one per county, which offered primary and emergency care, and sub-regional and regional hospitals supplied inpatient and specialized outpatient care (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). Any resident could receive care for free in its facilities, so coverage was universal (Alegría et al., 2001)[7] and free of cost for patients.

Puerto Rico was exempt from the federal requirement that Medicaid recipients be free to choose their providers, so its Medicaid program paid only for care in the state and municipal facilities and made no payments to private providers (Pagán-Berlucchi and Muse, 1983; Appendix E). Public health care coexisted with an extensive private sector. In 1990, the public system operated roughly 40 percent of hospital beds and served about 1.8 million residents, or 51.4% of the population (Instituto de Administración y Política de Salud, 1996; Appendix E). In 1990, Puerto Rico had 13 public and 56 private hospitals (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005).

In 1979, the Department of Health employed 57.3 percent of the island's practicing health professionals (Pagán-Berlucchi and Muse, 1983). In 1992, the island had 8,443 licensed physicians and 16,848 registered nurses, according to Junta de Planificación de Puerto Rico (2026) (Appendix Table A1). In 1993, the ambulatory health care provider industry employed 15,822 workers (see Appendix Table A2), and their county mean of annual pay per worker was $21,283 in 2025 dollars (see Table 1).

[7]See Appendix E for a description of the services each level provided and how patients were referred between levels.

Federal Medicaid funds, available in Puerto Rico beginning in 1965 (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005), financed part of public-system expenditures, and Puerto Rico covered the remainder. In fiscal year 1979, federal Medicaid payments, capped since 1968, covered 17.3 percent of public health system spending (Pagán-Berlucchi and Muse, 1983). Act 72 of 1993 attributed fiscal pressures to insufficient budgets, rising technology and supply costs (Commonwealth of Puerto Rico, 1993). The reform sought to reduce costs for the government through contracts with insurers and providers while preserving medical and hospital access for low-income patients (Commonwealth of Puerto Rico, 1993).

2.2 Reform implementation

Act 72 of September 7, 1993, implemented the health reform that reorganized public health care around Medicaid managed care (Commonwealth of Puerto Rico, 1993), on a rollout across different counties from 1994 to 2000. It created the Puerto Rico Health Insurance Administration to administer coverage funded through Medicaid and contract with health maintenance organizations (HMOs) (Centers for Medicare and Medicaid Services, 2000)[8]. In each county it reached, (i) the county's CDT and the public hospitals serving it passed to private providers through sales and long-term leases, (ii) contracts with private managed-care plans replaced government salaries as the way providers were paid, (iii) universal eligibility for care at the public facilities was replaced by self-enrollment in the government health plan for residents below 200 percent of the Federal poverty line[9], and (iv) free services were replaced by income-dependent cost sharing that ranged from $1 to $4 per visit and $0.50 to $3 per prescription (Instituto de Administración y Política de Salud, 1996; World Health Organization and Pan American Health Organization, 1998). Private MCOs managed

[8] Appendix B reports the reformed system's funding by source, including federal Medicaid.

[9] As of June 30, 2000, additional Medicaid groups eligible for the plan were children and adults in the Section 1931 (AFDC/TANF) and blind or disabled categories, the aged, children in foster care, children covered by CHIP (Title XXI), and police officers (Centers for Medicare and Medicaid Services, 2000).

provider networks and paid for covered services, and private providers delivered care.

The government paid insurers capitated premiums, which consisted of predetermined amounts per enrollee for specified services over a fixed period. Insurers then negotiated payment terms with network providers (Commonwealth of Puerto Rico, 1993; World Health Organization and Pan American Health Organization, 1998; Appendix E). Insurers also cut costs by making primary-care providers gatekeepers to specialized care, since specialist care, diagnostic tests, and procedures required their referrals (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). The government also sold the 78 CDTs and eleven hospitals or transferred them under long-term leases, while the island retained its two largest public hospitals (Instituto de Administración y Política de Salud, 1996; Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005; Appendix E)[10]. The government health plan offered broad benefits and low cost sharing to its beneficiaries. The benefit package covered hospital, ambulatory, pharmacy, dental and mental-health care without pre-existing-condition exclusions (Centers for Medicare and Medicaid Services, 2000), and Appendix B lists the covered services.

The government selected MCOs through competitive bidding (World Health Organization and Pan American Health Organization, 1998; Brugueras Fabre, 2002). The reform was implemented in the first counties in February 1994 and in San Juan in 2000 (Brugueras Fabre, 2002; see Figure 2). The six cohorts comprise 18 counties in 1994, 27 in 1995, 7 in 1996, 13 in 1997, 12 in 1998, and San Juan alone in 2000. Enrollment rose from 130,392 residents in 1994 to 1,795,067, or 47% of the population, when the rollout ended in 2000, out of 1,866,965 residents eligible that December (Administración de Seguros de Salud de Puerto Rico, 2026). By 2003, 7 percent of residents lacked a health plan, and the commission that evaluated the health system in 2005 attributes the growth of the uninsured population to government facilities that shrank or disappeared (Comisión para Evaluar el Sistema de Salud del Estado

[10]Fabre and Straub (2023) include long-term leases among partnership contracts and classify a sale, which transfers title, as privatization. The facility transfers therefore fall on both sides of their definition.

Libre Asociado de Puerto Rico, 2005). Appendix E reports take-up and the composition of coverage.

Enrollment at the end of the rollout was close to the 1.8 million residents the public system served in 1990 (Table 2), so the number of residents whose care the government financed changed little over the rollout. Cost sharing was low, so the reform did not rely on the prices patients paid to limit their use of care. Copayments and premiums were set administratively, so prices did not adjust to clear the market, and MCOs rationed any excess demand through referrals.[11]

Beyond the public facilities it transferred, the reform affected providers that were already private through the MCO's insurance networks. By joining one, a provider gained the plan's enrollees, mostly residents with incomes below 200 percent of the Federal poverty line. Whether the average needs of its patients rose or fell therefore depended on whether these enrollees needed more or less care than the privately insured, Medicare and public-employee patients it already served (Appendix E).

In the health labor market, the government, which in 1979 had employed most of the island's health professionals, reduced positions in its health delivery and regulatory agencies from 26,905 in fiscal year 1996 to 18,062 in 1999 (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1998). By 2000, private ambulatory employment had risen to 24,815 jobs and the county mean of annual pay per worker to $23,742 (Appendix Table A2; Table 1).

[11] I observe neither the payment terms MCOs negotiated with providers nor each plan's provider network, so I cannot test whether the plans changed which services patients could use or how often they sought care. The evidence I have points to little change. In 2003, 7 percent of residents lacked a health plan, against none before the reform (Table 2); the household share of consumption spent on medical care kept its pre-reform trend (Appendix Figure A1); and cost sharing went from nothing to $1 to $4 per visit, set by income, against a federal minimum wage of $5.15 an hour in 2000 (U.S. Department of Labor, Wage and Hour Division, 2026). Whether the reform changed patients' use of care therefore remains a limitation of this paper.

3 Data

3.1 Health outcomes

The Puerto Rico Planning Board provides annual population, births, all-cause and cause-specific deaths, and county characteristics (Junta de Planificación de Puerto Rico, 2026)[12]. The all-cause mortality panel has all 78 counties in every year from 1990 to 1999, and the tables note where a cause or a birth outcome is missing in a year. Table 1 summarizes the main outcomes, and Appendix Table A3 reports baseline characteristics by cohort.

All-cause and cause-specific mortality are deaths per 100,000 residents, without age adjustment, using annual intercensal population from the Planning Board in Puerto Rico and SEER on the mainland. Infant mortality is infant deaths per 1,000 live births, and maternal mortality is maternal deaths per 100,000 live births. The birth outcomes are shares of resident live births, as the vital-statistics reports define them: early prenatal care is care that began in the first trimester of pregnancy, and low birth weight is a weight below 2,500 grams regardless of gestational age, a category that includes very low birth weight, below 1,500 grams (Puerto Rico Department of Health, 2025). The government-hospital birth share is births in a government-operated hospital over all births. Monetary outcomes use constant 2025 dollars and the all-items consumer price index (U.S. Bureau of Labor Statistics, 2026). In 1993, the year before the reform, 99.8 percent of births on the island occurred in a hospital (Puerto Rico Department of Health, 2025). Because nearly every birth took place in a hospital, the ownership of the hospital where a birth occurred tracks the shift in supply from public to private providers. Patients rarely needed to leave their cohort for care. Every cohort had at least one public and one private hospital (Appendix Figure A2, panel (b)), and in 2000, 70.7 percent of resident workers were employed in a county of their own cohort.

[12]Digitized annual vital-statistics reports for 1990–2002 add births by hospital ownership, prenatal care, birth characteristics, and early observations for mortality causes incompletely recorded by the Planning Board (Puerto Rico Department of Health, 2025). I retain source flags and leave unobserved counts missing. Overlapping series have correlations of 0.99–1.00. The data-construction appendix documents the transcription checks, their exceptions, and classification changes.

That year, 85.6 percent of deaths occurred in a hospital or at home (Puerto Rico Department of Health, 2025), so county mortality reflects access to hospital and preventive care where residents live.

3.2 Death records

The Demographic Registry's death certificates record granular information on the Planning Board's county-level mortality statistics. They show the age and sex of each person who died. The Puerto Rico Institute of Statistics publishes them as individual records with the dates of death and birth, sex, marital status, and county of residence (Instituto de Estadísticas de Puerto Rico, 2026). The published table holds a random sample of 50 percent of the registry's records for 1990–2000. Since the records hold about half of the deaths, I weight every record by 1/0.5055, so that each stands for about two deaths and rates computed from the records are comparable with the Planning Board's. Because the weight is the same for every record, it scales the level of a rate and leaves shares and effects relative to the pre-treatment mean unchanged. Appendix Table A4 describes the records by year and breaks the deaths down by age and sex, as counts, as shares of the year's deaths, and as rates per 100,000 residents.

3.3 Healthcare supply

The Quarterly Census of Employment and Wages (QCEW) records annual employment, establishments, and payroll across the United States and Puerto Rico (U.S. Bureau of Labor Statistics, 2025). It counts jobs covered by unemployment insurance at their workplace, excluding self-employment. I measure county private health care supply in Ambulatory Health Care Services (NAICS 621): physicians' offices, outpatient clinics, and diagnostic services (Appendix Table A2, Panel A), so staff of transferred CDTs whom private providers kept enter observed employment when their facility is transferred. The QCEW panel has the

59 counties published in every year from 1990 to 2000. Their employment is 97.9 percent of the island's ambulatory employment in 1993 and 96.8 percent in 2000, measured as the panel counties' employment over the island total for the same year (Appendix Table A2, Panel C; Table 1).[13] I use NAICS 621 rather than the whole health care and social assistance sector (NAICS 62) for three reasons: ambulatory care is the care the capitated contracts paid for and the CDTs provided; NAICS 62 adds social assistance (NAICS 624), which is not healthcare; and NAICS 621 is published as one series, while a total of NAICS 621 to 623 sums parts that are not published in every county and year. The wider sector appears in County Business Patterns, whose health services series includes hospitals. The outcomes are employment and establishments per 1,000 residents, employees per establishment, and annual payroll per worker in constant 2025 dollars. The Centers for Medicare and Medicaid Services Provider of Services files identify hospitals certified for Medicare, the federal health insurance for older people and certain people with disabilities (Centers for Medicare and Medicaid Services, 2025b). The 1991–2005 panel covers all 78 counties, including 47 with no certified hospitals during this period, and the outcome is certified hospitals per 1,000 residents. In 1991, the first year of the files, they list 79 hospitals on the island, 4 of them federal. The other 75 are at least as many as the 69 public and private hospitals that Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico (2005) counts in 1990, so this dataset covers all of the island's public and private hospitals[14]. Fifty-five of the 79 participate in Medicare, 44 private hospitals and 11 government ones, and the government count reaches 13 by 1993. The other 24 are emergency hospitals, which do not participate in Medicare and can bill it only for emergency care (Centers for Medicare

[13]Two sources assess QCEW coverage. County Business Patterns closely matches pre-reform private ambulatory employment in the available county cross-section (U.S. Census Bureau, 2000b; Appendix Table A2, Panel D, Appendix Figure A3). Census microdata count more health care workers, with a narrowing gap over the decade (Ruggles et al., 2025; Appendix Table A2, Panel E). That comparison is island-wide and measures resident workers rather than workplace jobs. The data-construction appendix details differences in industry definitions, reporting, and reference periods.

[14]The difference between 1990's 69 hospitals and 1991's 79 hospitals could be the 4 veterans hospitals, in addition to the 3 psychiatric hospitals (if Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico (2005) counted general hospitals only), emergency-hospital registrations they do not count as hospitals, or hospitals that opened between 1990 and 1991.

and Medicaid Services, 2025b), and all 4 federal hospitals are among them. The files record no ownership or hospital type for the 24 and list the same ones in every year from 1991 to 2000, so every change in the count comes from hospitals that participate in Medicare.

3.4 Mainland comparison data

To extend comparisons beyond Puerto Rico's completed rollout, I assemble mainland county mortality from CDC WONDER (National Center for Health Statistics, 2026), population and age composition from Surveillance, Epidemiology, and End Results (SEER) (National Cancer Institute, 2026), and poverty and land area from the 1990 Census (U.S. Census Bureau, 1990). The primary sample excludes mainland states adopting Medicaid managed care by 2000. Section 6.3 uses them as a comparison group for robustness purposes.

4 Empirical strategy

4.1 Estimand and identification

I estimate the effects of the reform using the staggered difference-in-differences framework of Callaway and Sant'Anna (2021). For each adoption cohort, I compare changes in outcomes with those in counties not yet treated. Let Y_{it} denote county i's outcome in year t, G_i its adoption year, and D_{it} an indicator equal to one once county i has been treated by year t. The potential outcomes $Y_{it}(g)$ and $Y_{it}(0)$ correspond to first adopting in year g and to remaining untreated throughout. The group-time average treatment effect is $\mathrm{ATT}(g,t) = \mathbb{E}[Y_{it}(g) - Y_{it}(0) \mid G_i = g]$. Write $\bar{g}$ for the last adoption year and $\mathcal{G} = \mathrm{supp}(G) \setminus \{\bar{g}\}$. Because no county remains untreated throughout, $\mathcal{G}$ excludes the last-treated cohort, which has no valid untreated comparison group.

Without covariates, Callaway and Sant'Anna (2021) show that the group-time effect is

identified as

$$\mathrm{ATT}(g,t) = \mathbb{E}[Y_{it} - Y_{i,g-1} \mid G_i = g] - \mathbb{E}[Y_{it} - Y_{i,g-1} \mid D_{it} = 0]\,, \qquad g \in \mathcal{G}, \quad g \le t < \bar{g}, \tag{1}$$

where the second expectation is taken over counties not yet treated by year t, that is, those with $G_i > t$. Two assumptions deliver equation (1). Absent the reform, mortality in counties that entered in year g would have changed between $g-1$ and t by as much as mortality in counties not yet treated by t, and the same holds for every other outcome. Entry also changed nothing before it happened, so no outcome moved in anticipation. A county's cohort year is the year the reform reached it (Section 2.2), so $\mathrm{ATT}(g,t)$ is the effect of the reform's arrival together with the facility, payment and eligibility changes as they reached the county by year t.

The last county entered in 2000, so $\bar{g} = 2000$. By 1998 the other 77 counties had entered, leaving San Juan as the only comparison county in 1998 and 1999. I therefore identify effects for the earlier cohorts through 1999 and write $\bar{t} = \bar{g} - 1 = 1999$ for the last identified year. San Juan is only a comparison county, since its own post-adoption effect is not identified. Cohorts differ significantly at baseline, mostly in population (Appendix Table A3). The estimator of Callaway and Sant'Anna (2021) offers many advantages. It compares each cohort only with counties not yet treated, so a county that has already entered is never used as a control. Additionally, its doubly robust version also lets the untreated trend depend on baseline characteristics such as population, and the robustness specifications in Section 6 use it to account for the cohorts' differences.

4.2 Aggregation and inference

For each specification I also report an event study. Let $e = t - g$ denote event time, the number of years elapsed since a county adopted the reform. Summarizing how effects vary with length of exposure therefore amounts to averaging the $\mathrm{ATT}(g,t)$ across cohorts at a

common value of e, which is the event-study aggregation of Callaway and Sant'Anna (2021),

$$\theta_{es}(e) = \sum_{g \in \mathcal{G}} \mathbf{1}\{g + e \leq \bar{t}\} \; \mathrm{P}(G = g \mid G + e \leq \bar{t}) \; \mathrm{ATT}(g, g + e). \tag{2}$$

This is the average effect e years after adoption across the cohorts ever observed at that time period, and $e = 0$ gives the effect of the policy in the adoption year itself. The indicator restricts the sum to identified group-time effects (Callaway and Sant'Anna, 2021). The weights are cohort shares among the cohorts observed at event time e, so the set of cohorts behind $\theta_{es}(e)$ changes with e: the longest horizons rest on the 1994 and 1995 waves, and the earliest leads on the latest cohorts. Every group-time effect, before adoption as well as after, is measured from year $g - 1$, the year before the cohort adopted.

The baseline includes no covariates. Additional specifications in Section 6 use the doubly robust estimand of Callaway and Sant'Anna (2021), which allows untreated trends to depend on pre-reform county characteristics. The reported overall coefficient is θ^O_W in their notation, a weighted average of all identified group-time effects with weights proportional to cohort size, so counties contributing more post-reform years receive greater total weight. Inference follows with county-specific bootstrap multipliers and 1,000 replications.

5 Results

I first consider who delivered care, then the effects on health outcomes, on health care supply, on mortality by cause, and on mortality by age and sex.

5.1 Who delivered care

Relative to counties not yet treated, the government-hospital share of residents' live births fell by 17.94 percentage points (s.e. 1.25), 28.0 percent of its pre-treatment mean, and the

private-hospital share rose by 17.29 points (s.e. 1.19), 47.5 percent of its mean (Table 3; Figure 3). The proportional shift in patient utilization from public to private hospitals confirms that the reform replaced public hospitals with private ones.

5.2 Health outcomes

The all-cause rate counts residents' deaths. In 1993, the year before the reform, 54.5 percent of deaths occurred in a hospital, 30.5 percent in a government hospital and 24.0 percent in a private one, and 63.6 percent of the deceased were reported to have received medical assistance (Puerto Rico Department of Health, 2025). By 2000, as government hospitals passed to private management, 33.5 percent of deaths occurred in private hospitals, 22.1 percent in government hospitals and 30 percent at home (Puerto Rico Department of Health, 2025). Homicides and accidents were 3.3 and 4.2 percent of deaths in 1993 (Appendix Table A5). Assuming that home deaths are related to preventable health complications, the reform could have affected the death rate through the care patients received in hospitals or through the preventive health care received before passing away at home.

I estimate an increase in all-cause mortality of 26.96 additional annual deaths per 100,000 residents (s.e. 10.32), equivalent to 3.7 percent of the pre-treatment mean of 724.2 (Table 4).[15] In the event study (Figure 4), the effect grows with exposure, and every lead but the earliest, eight years before entry, has an interval that includes zero. Since a little more than half of the deceased died in a hospital, and the remainder mostly died at home, the 3.7 percent rise in the mortality rate can plausibly be an effect of the reform's reorganization on health care supply.

To understand the effect the health reform had on preventive health care, I look at early prenatal care. I interpret it as a proxy for utilization of preventive health care, which cost-cutting efforts can reduce. The share of births with early prenatal care falls by 2.33

[15] Appendix Table A5 reports island deaths by cause and year.

percentage points (s.e. 0.62), or 3.0 percent of its pre-treatment mean (Table 5; Figure 5). The low-birth-weight share rises by 0.67 percentage points (s.e. 0.22), or 6.8 percent of its baseline share. The very-low-birth-weight estimate is also positive, at 6.7 percent of its baseline mean, but is not statistically distinguishable from zero.[16] Infant and maternal mortality estimates are also statistically indistinguishable from zero. Duggan (2004) likewise finds no detectable effect of California's Medicaid managed-care mandates on in-hospital mortality among Medicaid-insured infants. But the fall in early prenatal care and the rise in low birth weight are consistent with the findings of Aizer et al. (2007) and indicate reduced access to preventive health care or lower treatment intensity.

5.3 Healthcare supply

The health care supply estimates measure slower growth. Ambulatory health care employment per 1,000 residents rose in the average county of the QCEW panel from 2.24 in 1990 to 4.83 in 2000 (Table 1), so a negative estimate means that counties grew less after the reform reached them than counties it had not yet reached.

Ambulatory employment runs 0.2663 jobs per 1,000 residents below its counterfactual path (s.e. 0.1494), 9.6 percent of the pre-treatment mean of 2.773 and significant at the 10 percent level, and establishments per 1,000 residents are 0.0490 lower (s.e. 0.0185), 6.5 percent of 0.7494 (Table 6). Annual wages per worker are $704 lower (s.e. 239), 3.4 percent of $20,887 in constant 2025 dollars, and hospitals per 1,000 residents are 0.0006 lower (s.e. 0.0002), 3.4 percent of their pre-treatment mean. Figure 6 reports the event studies. Most leads are close to zero, but the intervals exclude zero for employment eight and seven years before entry, establishments seven years before, and certified hospitals six years before.

Employees per establishment rise by 0.2543 (s.e. 0.1511), 7.6 percent of 3.356, an estimate significant at the 10 percent level. The ratio is not a linear function of the two per-resident

[16] Appendix Table A6 reports annual birth outcomes for all 78 counties.

outcomes, so its estimate need not follow from theirs, which at the pre-treatment means imply a ratio about 3 percent lower. It divides county employment by county establishments, so it rises either when establishments grow, through consolidations or mergers, or when small establishments close, even if no establishment grows.

These estimates describe ambulatory health care (Section 3.3), which includes primary and preventive care. Employment and wages per worker both below their counterfactual paths show a decrease in the demand for ambulatory health care workers. At the old wage, more workers sought ambulatory health care jobs than employers wanted to hire, and the wage fell until that surplus cleared. In the framework of Appendix C, a shortfall of this kind is consistent with cost-reducing innovation and with labor-saving quality innovation alike. Its direction matches the lower staffing per patient at privatized U.S. public hospitals (Duggan et al., 2023). Together with higher mortality, the results are consistent with cost containment whose harm to patients outweighed any quality gain, the case in which Proposition 1 predicts higher mortality under private control.

5.4 Mortality by cause

This subsection reports the effect on mortality from the most common causes of death, twelve named causes and a residual of all other deaths, each measured as deaths per 100,000 residents (Table 7; Figure 7).[17] Each cause can depend on several services and on factors outside healthcare, so a cause-specific estimate does not measure the quality of one corresponding service.

Cardiovascular and metabolic causes show mixed effects. Diabetes mortality increases

[17]Appendix Table A7 reports the two-way fixed effects estimate for each cause, with log population, the unemployment rate, crime and the share of residents receiving benefits from the Nutrition Assistance Program (NAP) as control variables, each measured in the same year as mortality. In place of the federal SNAP entitlement, Puerto Rico receives federal food aid through NAP, a fixed block grant. The estimates for the eight causes recorded in every year and for the residual of all other deaths sum to the estimate for all-cause mortality with the same control variables (Section 6.4). The table shows how each cause's estimate relates to the TWFE estimate for all-cause mortality. Chronic pulmonary disease has the largest estimate in that table.

by 4.913 deaths per 100,000 (s.e. 2.409), or 9.5 percent of its pre-treatment mean, whereas hypertension mortality falls by 5.943 (s.e. 2.221), or 22.3 percent. Both conditions can benefit from regular monitoring and medication, but these opposing estimates do not support a general claim that chronic disease management improved or deteriorated. Heart-disease and cerebrovascular mortality estimates are 1.009 (s.e. 3.980) and −1.285 (s.e. 2.306), respectively; neither is statistically distinguishable from zero.

Among respiratory and infectious causes, AIDS mortality increases by 10.22 deaths per 100,000 (s.e. 2.177), or 41.9 percent. Antiretroviral treatment is linked to lower mortality (Palella et al., 1998), making treatment access a potential mechanism. Establishing that the reform disrupted this access, however, requires evidence on treatment receipt and continuity. The estimates for chronic pulmonary disease, pneumonia and influenza, and septicemia are positive but statistically indistinguishable from zero.

Cancer and chronic liver disease show no common pattern. Cancer mortality falls by 9.277 deaths per 100,000 (s.e. 4.245), or 8.1 percent, but attributing that reduction to earlier diagnosis or improved treatment would require evidence on disease stage, treatment receipt, and survival following diagnosis. The cirrhosis estimate is −0.644 (s.e. 1.245) and statistically indistinguishable from zero.

External causes raise the question of timely emergency care. Homicide mortality increases by 5.766 deaths per 100,000 (s.e. 1.974), or 41.8 percent, while the accident estimate is −1.533 (s.e. 2.433) and statistically indistinguishable from zero. Timely access to an appropriately equipped facility can matter for survival after severe injury, so the homicide estimate may also reflect reduced emergency capacity: with fewer hospitals and ambulatory establishments than without the reform, patients needing immediate care could have traveled farther. However, testing this hypothesis requires data on emergency transport and treatment.

The residual, all deaths less the twelve named causes, has the largest positive estimate, 20.60 deaths per 100,000 (s.e. 7.030), or 10.8 percent of its pre-treatment mean of 190.0. The

increase suggests that the reform's adverse mortality effects extended beyond the named causes, but the residual mixes conditions with different care needs, so the estimate cannot say whether the increase was concentrated in a few conditions or spread across many. Mixed results across these different causes of death suggest that the health reform affected the provision of different health care services in different ways. Finer cause-of-death and healthcare-use data would be needed to determine whether reduced access, treatment disruptions, or other mechanisms explain the increase.

5.5 Mortality by age and sex

The Demographic Registry's records give the age and sex of each person who died (Section 3.2), so the all-cause estimate can also be decomposed by who died (Table 8; Figure 8). As with the causes, each column counts the deaths of one group per 100,000 residents and compares them between treated counties and counties not yet treated, so the three age groups add up to the all-cause estimate from the records in column 1, as do men and women, apart from the few deaths of unknown age or sex.

Since the health reform operated through Medicaid MCOs, I look at whether working-age adults, who are ineligible for Medicare or CHIP but potentially eligible for Medicaid, account for the increase in the mortality rate. Deaths of residents aged 18 to 64 (a group whose deaths accounted for 33 percent of all deaths; Appendix Table A4) rise by 25.48 per 100,000 residents (s.e. 7.668), 11.2 percent of the group's pre-treatment mean of 227.5, more than the all-cause estimate from the records, 23.67 (s.e. 12.43). Among residents aged 65 and over, the Medicare-eligible population, who accounted for about 62 percent of deaths, the estimate is -6.438 (s.e. 10.59), -1.4 percent of the group's pre-treatment mean of 468.0; for those under 18 (the CHIP-eligible population) it is 3.723 (s.e. 2.488). Neither is statistically distinguishable from zero. The increase thus falls among working-age adults and not in the Medicare- or CHIP-eligible populations; the reform most directly affected residents without

private insurance, Medicare, CHIP or coverage through the public employees' insurance plan (Table 2).

By sex, the increase is among men: 24.23 additional deaths per 100,000 residents (s.e. 10.60), 5.6 percent of their pre-treatment mean and significant at the 5 percent level, as large as the all-cause estimate from the records, against −0.5675 (s.e. 6.450) for women. Men accounted for 59 percent of deaths in 1990–1993 (Appendix Table A4). The concentration among men is consistent with the causes that rose most. For example, the largest estimate among the named causes of death indicates that deaths from AIDS rose by 41.9 percent of their pre-treatment mean, and in 1994, Puerto Rico Department of Health (2025) records 1,549 deaths from AIDS in Puerto Rico, 42.0 per 100,000 residents. Of these, 1,210 were men, 67.8 per 100,000 men, and the 1,164 deaths of men aged 20 to 64 alone come to 65.2 per 100,000 men. This further supports the hypothesis that the reform may have decreased access to healthcare related to AIDS treatment.

6 Robustness

6.1 Covariate-specific trends

Rollout timing may be related to county characteristics that affect mortality trends. I therefore allow for covariate-specific trends. I control for baseline covariates using the doubly robust estimator of Callaway and Sant'Anna (2021), which lets untreated mortality trends depend on the baseline characteristics on which the adoption cohorts differ significantly (Appendix Table A3): the 1990–1993 averages of log population, the share of residents who are beneficiaries of the Nutrition Assistance Program (NAP), the share of births in government hospitals, the unemployment rate, the crime rate, and the 1990 share of residents aged 65 and over (i.e., residents with high health care utilization) (Table 9; Figure 9).

After including these covariates, results remain consistent. Conditioning on one charac-

teristic at a time leaves the estimate between 22.75 and 28.15 additional deaths per 100,000 residents (Table 9, columns 2 to 7; Figure 9), all of them significant at the 5 percent level. With all six together, the estimate is 15.09 (s.e. 13.70), 2.1 percent of the pre-treatment mean, and not statistically distinguishable from zero (column 8). That column rests on 2.6 effective comparison counties, so few comparison counties resemble the treated ones in all six covariates, and its lack of statistical significance could reflect the resulting imprecision.

6.2 Sample, weights and scale

Table 10 changes the comparison group, the weights, and the scale, and Figure 10 reports the event studies. Appendix Table A8 reports the same kinds of specification for the causes of death and the birth outcomes.

Counting deaths in the Demographic Registry's records rather than the Planning Board's counts gives 23.67 additional deaths per 100,000 residents (s.e. 12.43), 3.2 percent of the pre-treatment mean, close to the baseline but less precise because the records hold half the deaths (Table 10, column 2).

Weighting counties by their 1990–1993 population gives more influence to the populous counties, where provider concentration and reliance on public care differ, and less to small counties, whose mortality rates are more volatile. The weighted estimate is 32.27 additional deaths per 100,000 residents (s.e. 8.408), or 4.5 percent of the pre-treatment mean, against 3.7 percent in the baseline (Table 10, column 3), so the increase keeps its sign and significance when counties count by population.

Rates can change because deaths change, because population changes, or both, which matters if the reform coincided with people's migration or it pushed people to move. In death counts, the estimate is 173.2 additional deaths per county-year (s.e. 9.233), or 54.2 percent of the pre-treatment mean of 319.3 (Table 10, column 4), but this number does not estimate the reform's effect on deaths. In its event study (Figure 10), deaths in counties the

reform reached were already rising relative to counties not yet treated from eight years before it reached them, so the post-entry estimates combine any effect of the reform with that earlier trend. Death counts grow with a county's population, so counties whose populations grow at different rates add deaths at different rates even when their death rates move together; dividing by population removes that difference, which is why the main outcome is the rate.

In logs, the estimate is 0.0247 (s.e. 0.0145), about 2.5 percent, smaller than the baseline and significant only at the 10 percent level (Table 10, column 5).[18] The baseline's 3.7 percent divides the effect in levels by the pre-treatment mean, which weights each county's proportional change by its mortality, while the log estimate averages proportional changes. Therefore, the gap is consistent with larger proportional effects in counties with higher mortality.

6.3 Mainland comparison

I also use the mainland counties of Section 3.4 as a comparison group. These counties are never treated and cannot receive spillovers from treated counties on the island, such as those from daily commuting. Additionally, this comparison allows the estimate to reflect the effects in 2000, including San Juan's adoption-year effect. The doubly robust estimator of Callaway and Sant'Anna (2021) lets untreated trends depend on 1990–1993 population density and the 1990 poverty rate. With 282 comparison counties, the estimate through 2000 is 23.15 additional deaths per 100,000 residents (s.e. 20.54), or 3.2 percent of the pre-treatment mean of 731.8 (Table 10, column 6), close to the baseline estimate but statistically indistinguishable from zero.[19]

[18] The transformation is $\log(1 + m_{it})$, where m_{it} denotes deaths per 100,000 residents. The conversion $100[\exp(0.0247) - 1]$ equals 2.5 percent on the multiplicative scale for $1 + m_{it}$, and because mortality averages 724.2 deaths per 100,000 before treatment, it is also the percent change in mortality.

[19] Alternative mainland samples and estimates through 2003 are also positive and imprecise. With 353 counties in states without Medicaid health maintenance organization (HMO) mandates, the type of MCO Puerto Rico contracted (Centers for Medicare and Medicaid Services, 2000), the estimate through 2000 is 21.57 deaths per 100,000 (s.e. 21.67). Through 2003, the non-MMC and non-HMO estimates are 16.22 (s.e. 19.39) and 14.93 (s.e. 27.23), or 2.2 and 2.0 percent of the same baseline mean (Appendix Table A9;

6.4 Alternative estimators

Table 11 reports five estimators, four of which allow treatment effects to vary across cohorts and time. The baseline estimator of Callaway and Sant'Anna (2021) yields 26.96 additional deaths per 100,000 residents (s.e. 10.32), against 31.08 (s.e. 9.090) for Sun and Abraham (2021). The extended two-way fixed effects estimator of Wooldridge (2025) and the imputation estimator of Borusyak et al. (2024) both yield 37.08, with standard errors of 7.533 and 7.320. All four are statistically significant at the 1 percent level and correspond to 3.7 to 5.1 percent of the pre-treatment mean, so the baseline is the most conservative of the four (Figure 11 compares their event studies).

The conventional two-way fixed effects (TWFE) regression instead yields −1.283 deaths per 100,000 residents (s.e. 10.20), statistically indistinguishable from zero.[20] With staggered adoption, TWFE uses counties already treated as comparisons for counties treated later, so changes in the treatment effects of those comparison counties enter its estimate (Callaway and Sant'Anna, 2021; Borusyak et al., 2024). Following de Chaisemartin and D'Haultfœuille (2020), TWFE here places negative weight on 32.3 percent of treated county-years, and the negative weights sum to 16.3 percent of the positive ones in absolute value (Appendix Table A10). The TWFE coefficient therefore need not recover the average treatment effect, which is why the baseline uses Callaway and Sant'Anna (2021) and TWFE is a diagnostic here.

Appendix Figure A4).

[20] Appendix Table A10 reports TWFE without covariates and with log population, the unemployment rate, crime and the NAP recipient share, each measured in the same year as mortality, one at a time and together. The six estimates range from −1.756 (s.e. 9.907), with crime, to 5.618 (s.e. 9.899), with log population, or −0.2 to 0.8 percent of the pre-treatment mean, and none is statistically distinguishable from zero.

7 Conclusion

This paper examines how Puerto Rico moved its public health care from public to private management. The baseline estimates indicate that the reform increased all-cause mortality by 26.96 annual deaths per 100,000 residents, equivalent to 3.7 percent of the pre-treatment mean, relative to the counterfactual. The estimate stays positive and significant for working-age adults, when I condition on the counties' baseline characteristics one at a time, when counties are weighted by population, and under three other difference-in-differences estimators that allow effects to vary across cohorts and years. It is positive but indistinguishable from zero when all six characteristics enter together and against mainland counties.

I examined two mechanisms, slower growth in ambulatory health care supply and lower use of timely preventive health care. Both are consistent with the incomplete-contracts model's concern that stronger incentives to reduce costs can harm patients when relevant dimensions of health care are difficult to specify, measure, and enforce in contracts. This is also consistent with Fabre and Straub (2023), who find the results of the public-private partnerships (PPPs) they survey mostly disappointing and trace them in part to weak incentives for providers to improve quality.

Because the reform changed facility control, provider payment, and eligibility together, the estimates measure their joint effect. Future work could estimate the reform's fiscal savings and its net effect on welfare. Longer-run comparisons would help distinguish disruption from persistent effects; variation separating policy components would help explain which features affected patients the most.

The estimates speak most directly to publicly financed systems that still own and staff their facilities and increasingly rely on private providers. Examples include England, where the Department of Health and Social Care purchased £24.1 billion in care outside the National Health Service in 2024–25 (National Audit Office, 2026), and New Zealand, where the share of publicly funded elective procedures performed by private providers nearly doubled,

from 8.6 percent in 2011–12 to 16.7 percent in 2023–24 (Radio New Zealand, 2026). The closest current case in the United States of America is the Veterans Health Administration, which buys a growing share of its care from private providers. In fact, the Veterans Health Administration (which owns and staffs more than 1,700 sites) raised its spending on outsourced care from $7.9 billion in 2014 to $17.6 billion in 2021 (in 2021 dollars), going from about 12 to 20 percent of its medical-care spending (Panangala et al., 2025; Congressional Budget Office, 2021). Moving all of veterans' care to private providers is the nearest analogue to the transfer studied in this setting. Medicaid already pays private providers, and states already contract much of its care to private managed-care organizations (Centers for Medicare and Medicaid Services, 2000), so the estimates do not predict the effect of extending managed care further. Read in reverse, they may indicate what a move from private management of publicly financed care toward public provision would aim to recover, although further research is needed to conclude what undoing the transfer would do to reverse its effect.

Governments undertaking similar transitions should make evidence of patient health outcomes a central consideration in managed-care contract renewals, ensuring that cost savings are not achieved at the expense of patient health.

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I. Tables

Table 1: Descriptive statistics by year, Puerto Rico counties

Variable	1990	1992	1994	1996	1998	2000
Panel A: Mortality, births and population						
All-cause mortality per 100,000	703.5	720.2	728.4	763.5	767.2	718.3
	(116.7)	(103.9)	(120.9)	(126.5)	(114.3)	(113.2)
Government-hospital birth share (percent)	65.8	65.3	63.5	51.6	36.6	30.6
	(11.3)	(10.9)	(10.4)	(15.2)	(14.6)	(19.9)
Private-hospital birth share (percent)	33.8	34.4	36.1	48.1	63.1	69.2
	(11.3)	(11.0)	(10.6)	(15.3)	(14.6)	(19.9)
Population	45,228	45,884	47,253	47,240	48,053	48,828
	(59,205)	(60,064)	(61,281)	(58,898)	(58,930)	(59,001)
Unemployment rate (percent)	17.56	18.99	18.83	16.23	16.16	13.24
	(4.56)	(4.95)	(4.94)	(4.57)	(4.70)	(4.15)
Treated counties	0	0	18	52	77	78
Not-yet-treated counties	78	78	60	26	1	0
% of population observed	100.0	100.0	100.0	100.0	100.0	100.0
Panel B: Ambulatory Health Care Services (NAICS 621)						
Employment per 1,000 residents	2.24	2.75	3.20	3.71	4.47	4.83
	(2.15)	(2.79)	(3.02)	(2.96)	(3.59)	(3.56)
Establishments per 1,000 residents	0.670	0.765	0.782	0.887	0.990	1.089
	(0.473)	(0.515)	(0.518)	(0.576)	(0.616)	(0.657)
Employees per establishment	3.11	3.27	3.81	4.01	4.29	4.27
	(1.37)	(1.60)	(1.67)	(1.46)	(1.69)	(1.44)
Wages per worker (2025 USD)	20,104	20,821	22,031	22,762	23,702	23,742
	(5,470)	(5,251)	(4,945)	(4,238)	(4,678)	(4,729)
Treated counties	0	0	9	37	58	59
Not-yet-treated counties	59	59	50	22	1	0
% of population observed	90.7	90.7	90.6	90.6	90.5	90.5
% of employment observed	97.9	97.8	97.7	96.9	96.4	96.8
Panel C: Medicare-certified providers						
Hospitals per 1,000 residents		0.018	0.018	0.018	0.017	0.017
		(0.027)	(0.027)	(0.026)	(0.026)	(0.025)
Treated counties		0	18	52	77	78
Not-yet-treated counties		78	60	26	1	0
% of population observed		100.0	100.0	100.0	100.0	100.0

Notes: County means with standard deviations in parentheses. Percent of population observed is the population of the counties this panel covers in that year, divided by the population of every county in that year. Sources: Panel A, the Puerto Rico Planning Board's municipal series; Panel B, the Bureau of Labor Statistics Quarterly Census of Employment and Wages; Panel C, the Medicare Provider of Services file. Percent of employment observed is the employment of the counties this panel covers in that year over the island total for the same year; employees per establishment is county employment over county establishments; wages are annual payroll over average annual employment, in 2025 dollars; the government-hospital birth share is births in a government-operated hospital divided by all births.

Table 2: Coverage of residents, 1990 and 2003

	1990		2003	
Coverage	Millions	Percent	Millions	Percent
Private insurance	1.2	34	1.2	32
Medicare	0.3	9	0.5	14
Public employees' insurance	0.2	6	0.5	12
Public healthcare[a]	1.8	51	1.5	40
Uncovered	0.0	0	0.3	7
Counted twice[b]			−0.2	−5
Total residents	3.5	100	3.8	100

Notes: The 1990 counts are the Planning Board's, from its 1996 economic report as Pan American Health Organization (2007) reports them, rounded to 0.1 million; the percentages are of their 3.5 million total. The 2003 percentages are the commission's (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005), and the 2003 counts apply them to the U.S. Census Bureau's estimate of 3,825,628 residents in 2003, rounded to 0.1 million; the Health Insurance Administration counted 1.52 million enrollees in December 2003 (Administración de Seguros de Salud de Puerto Rico, 2026). Public employees enrolled in the government health plan are counted in it (Pan American Health Organization, 2007), and the sources do not say whether the public employees' figures include dependents. [a] In 1990, the public healthcare operated under the public delivery system. Then, in 2003, it operated under the government health plan administered by Medicaid MCOs. [b] About 5 percent of residents, eligible for both, are counted in the government health plan and in Medicare (Pan American Health Organization, 2007); this row removes the second count.

Table 3: Effect of the health reform on utilization between types of providers

	(1) Government-hospital births	(2) Private-hospital births
ATT	-0.1794***	0.1729***
	(0.0125)	(0.0119)
Pre-treatment mean	0.6418	0.3642
ATT as % of pre-treatment mean	-28.0%	47.5%
Treated counties	78	78

Notes: Average treatment effects on the treated are estimated with Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Outcomes are share of births. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 4: Effect of the health reform on all-cause mortality

	(1) All causes per 100,000
ATT	26.96***
	(10.32)
Pre-treatment mean	724.2
ATT as % of pre-treatment mean	3.7%
Treated counties	78

Notes: Average treatment effects on the treated are estimated with Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 5: Effect of the health reform on prenatal care and prenatal-care-sensitive outcomes

Panel A. Infant and maternal mortality

	(1) Infant mortality per 1,000 births	(2) Maternal mortality per 100,000 births
ATT	0.0985	-4.291
	(0.7941)	(22.43)
Pre-treatment mean	12.56	19.27
ATT as % of pre-treatment mean	0.8%	-22.3%
Treated counties	78	78

Panel B. Birth weight and prenatal care

	(3) Low birth weight share of births	(4) Very low birth weight share of births	(5) Early prenatal care share of births
ATT	0.0067***	0.0008	-0.0233***
	(0.0022)	(0.0010)	(0.0062)
Pre-treatment mean	0.0987	0.0121	0.7735
ATT as % of pre-treatment mean	6.8%	6.7%	-3.0%
Treated counties	78	78	78

Notes: Average treatment effects on the treated are estimated with Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Low birth weight, very low birth weight and early prenatal care are unavailable for 1992. Each covers all 78 counties in the years it appears. Dataset is unbalanced. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 6: Effect of the health reform on health supply

Panel A. Employment and establishments

	(1) Employment per 1,000 residents	(2) Establishments per 1,000 residents	(3) Employees per establishment ratio
ATT	-0.2663*	-0.0490***	0.2543*
	(0.1494)	(0.0185)	(0.1511)
Pre-treatment mean	2.773	0.7494	3.356
ATT as % of pre-treatment mean	-9.6%	-6.5%	7.6%
Treated counties	59	59	59

Panel B. Wages and certified hospitals

	(4) Wages per worker 2025 USD	(5) Certified hospitals per 1,000 residents
ATT	-703.9***	-0.0006***
	(238.9)	(0.0002)
Pre-treatment mean	20,886.5	0.0166
ATT as % of pre-treatment mean	-3.4%	-3.4%
Treated counties	59	78

Notes: Average treatment effects on the treated are estimated with Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Columns (1) to (4) come from the Bureau of Labor Statistics Quarterly Census of Employment and Wages and column (5) from the Medicare Provider of Services file. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 7: Effect of the health reform on mortality by cause

Panel A. All-cause mortality				
	(1)			
	All causes			
ATT	26.96***			
	(10.32)			
Pre-treatment mean	724.2			
ATT as % of pre-treatment mean	3.7%			
Treated counties	78			
Panel B. Cardiovascular and metabolic causes				
	(2)	(3)	(4)	(5)
	Heart disease	Cerebrovascular	Hypertension	Diabetes
ATT	1.009	-1.285	-5.943***	4.913**
	(3.980)	(2.306)	(2.221)	(2.409)
Pre-treatment mean	156.6	36.72	26.68	51.55
ATT as % of pre-treatment mean	0.6%	-3.5%	-22.3%	9.5%
Treated counties	78	78	78	78
Panel C. Respiratory and infectious causes				
	(6)	(7)	(8)	(9)
	Chronic pulmonary	Pneumonia and influenza	Septicemia	AIDS
ATT	1.766	0.9216	2.097	10.22***
	(2.205)	(2.363)	(1.362)	(2.177)
Pre-treatment mean	29.12	29.44	11.31	24.39
ATT as % of pre-treatment mean	6.1%	3.1%	18.5%	41.9%
Treated counties	78	78	78	78

Table 7: Effect of the health reform on mortality by cause (continued)

Panel D. Cancer and chronic liver disease		
	(10) Cancer	(11) Cirrhosis
ATT	-9.277**	-0.6440
	(4.245)	(1.245)
Pre-treatment mean	114.4	18.64
ATT as % of pre-treatment mean	-8.1%	-3.5%
Treated counties	78	78
Panel E. External causes		
	(12) Accidents	(13) Homicide
ATT	-1.533	5.766***
	(2.433)	(1.974)
Pre-treatment mean	36.02	13.78
ATT as % of pre-treatment mean	-4.3%	41.8%
Treated counties	78	78
Panel F. Unclassified		
	(14) Unclassified	
ATT	20.60***	
	(7.030)	
Pre-treatment mean	190.0	
ATT as % of pre-treatment mean	10.8%	
Treated counties	78	

Notes: Average treatment effects on the treated are estimated with Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Chronic pulmonary and homicide are unavailable for 1995; septicemia is unavailable for 1990, 1991, 1992, 1993 and 1995; AIDS is unavailable for 1990, 1991 and 1995. Each covers all 78 counties in the years it appears. Unclassified is all-cause deaths less the causes printed above it in the years each is recorded, so in a year a cause is unavailable it also holds that cause's deaths. Each cause in the table is the following share of the pre-treatment all-cause mean: heart disease 22%, cerebrovascular 5%, hypertension 4%, diabetes 7%, chronic pulmonary 4%, pneumonia and influenza 4%, septicemia 2%, AIDS 3%, cancer 16%, cirrhosis 3%, accidents 5%, homicide 2%, unclassified 26%. Outcomes are per 100,000. Dataset is unbalanced. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 8: Effect of the health reform on mortality by age and sex

	(1) All residents	(2) Under 18	(3) 18 to 64	(4) 65 and over	(5) Men	(6) Women
ATT	23.67*	3.723	25.48***	-6.438	24.23**	-0.5675
	(12.43)	(2.488)	(7.668)	(10.59)	(10.60)	(6.450)
Pre-treatment mean	729.5	33.85	227.5	468.0	430.2	299.3
ATT as % of pre-treatment mean	3.2%	11.0%	11.2%	-1.4%	5.6%	-0.2%
Treated counties	78	78	78	78	78	78

Notes: Average treatment effects on the treated are estimated with Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Deaths are the Demographic Registry's death records of residents, each weighted by 1/0.5055, the inverse of the share of the Planning Board's 1990 to 1999 deaths the records hold. Every column divides deaths by all residents. Age is age at death. The age columns, (2) to (4), add up to column (1) together with the 104 deaths of unknown age, whose ATT is 0.9071. The sex columns, (5) and (6), add up to column (1) together with the 2 deaths of indeterminate sex, whose ATT is 0.0065. Outcomes are per 100,000. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 9: Alternative specifications: Covariate-specific trends

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
All-cause mortality								
ATT (per 100,000)	26.96***	28.15***	27.26**	22.75**	26.01**	26.84**	26.52**	15.09
	(10.32)	(10.53)	(10.97)	(10.97)	(10.88)	(10.44)	(10.53)	(13.70)
Pre-treatment mean	724.2	724.2	724.2	724.2	724.2	724.2	724.2	724.2
% of pre-treatment mean	3.7%	3.9%	3.8%	3.1%	3.6%	3.7%	3.7%	2.1%
Covariates								
Log population	No	Yes	No	No	No	No	No	Yes
NAP recipient share	No	No	Yes	No	No	No	No	Yes
Government-hospital birth share	No	No	No	Yes	No	No	No	Yes
Unemployment rate	No	No	No	No	Yes	No	No	Yes
Age 65 and over share	No	No	No	No	No	Yes	No	Yes
Crime	No	No	No	No	No	No	Yes	Yes
Treated counties	78	78	78	78	78	78	78	78
Estimated through	1999	1999	1999	1999	1999	1999	1999	1999
Effective comparison counties	.	21.44	16.58	8.671	12.17	25.28	21.35	2.630
Group-time cells dropped	0	0	0	0	0	0	0	0

Notes: Average treatment effects on the treated are estimated with Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Each control variable is one number per county. 1990 to 1993 means: log population, NAP recipient share, government-hospital birth share, unemployment rate and crime. 1990 Census values: age 65 and over share. Log population is the natural log of the county's population; the NAP recipient share is residents receiving Nutrition Assistance Program benefits over county population; the government-hospital birth share is births in government-operated hospitals over all births; unemployment is the county annual rate; the age 65 and over share is the 1990 Census counts aged 65 to 84 and 85 and over, over total county population; crime is Type I offenses per 1,000 residents, murders included. Conditional columns reweight the comparison group by the estimated odds of cohort membership at each cohort's base period; that probability falls outside 0.05 to 0.95 in 80 county-cohort comparisons in column (8), where the weights are least even. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 10: Alternative specifications

	(1) Baseline	(2) Death records	(3) Population weighted	(4) Levels	(5) Logs	(6) Never-treated
All-cause mortality						
ATT	26.96***	23.67*	32.27***	173.2***	0.0247*	23.15
	(10.32)	(12.43)	(8.408)	(9.233)	(0.0145)	(20.54)
Pre-treatment mean	724.2	729.5	724.2	319.3	6.576	731.8
ATT as % of pre-treatment mean	3.7%	3.2%	4.5%	54.2%	2.5%	3.2%
Treated counties	78	78	78	78	78	78
Never-treated counties						282

Notes: Average treatment effects on the treated are estimated with Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Column (4) is deaths per county-year and column (5) is the log of one plus deaths per 100,000; every other column is deaths per 100,000 residents. Every column is the Callaway and Sant'Anna estimator. Column (2) replaces the Planning Board's deaths with the Demographic Registry's death records of residents, each weighted by 1/0.5055, the inverse of the share of the Planning Board's 1990 to 1999 deaths the records hold; the population is the same. The never-treated column is conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census), the same covariates as the never-treated tables. Non-MMC states had not adopted Medicaid managed care by the horizon. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10). 282 of them enter the estimate. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 11: Alternative estimators

	(1) Callaway & Sant'Anna	(2) TWFE	(3) Sun & Abraham	(4) ETWFE	(5) Borusyak, Jaravel & Spiess
All-cause mortality					
ATT	26.96***	-1.283	31.08***	37.08***	37.08***
	(10.32)	(10.20)	(9.090)	(7.533)	(7.320)
Pre-treatment mean	724.2	724.2	724.2	724.2	724.2
ATT as % of pre-treatment mean	3.7%	-0.2%	4.3%	5.1%	5.1%
Treated counties	78	78	78	78	78

Notes: TWFE is two-way fixed effects OLS and ETWFE is the extended two-way fixed effects estimator of Wooldridge (2025). The named columns are Callaway and Sant'Anna (2021), Sun and Abraham (2021), and the imputation estimator of Borusyak, Jaravel and Spiess (2024). Callaway and Sant'Anna is the baseline identification strategy. Standard errors in parentheses: for CS, county-clustered multiplier bootstrap, 1,000 replications; for all others, clustered by county. Puerto Rico counties, 1990 to 1999. For Callaway and Sant'Anna, the control group is the not-yet-treated counties. The outcome is measured per 100,000. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

II. Figures

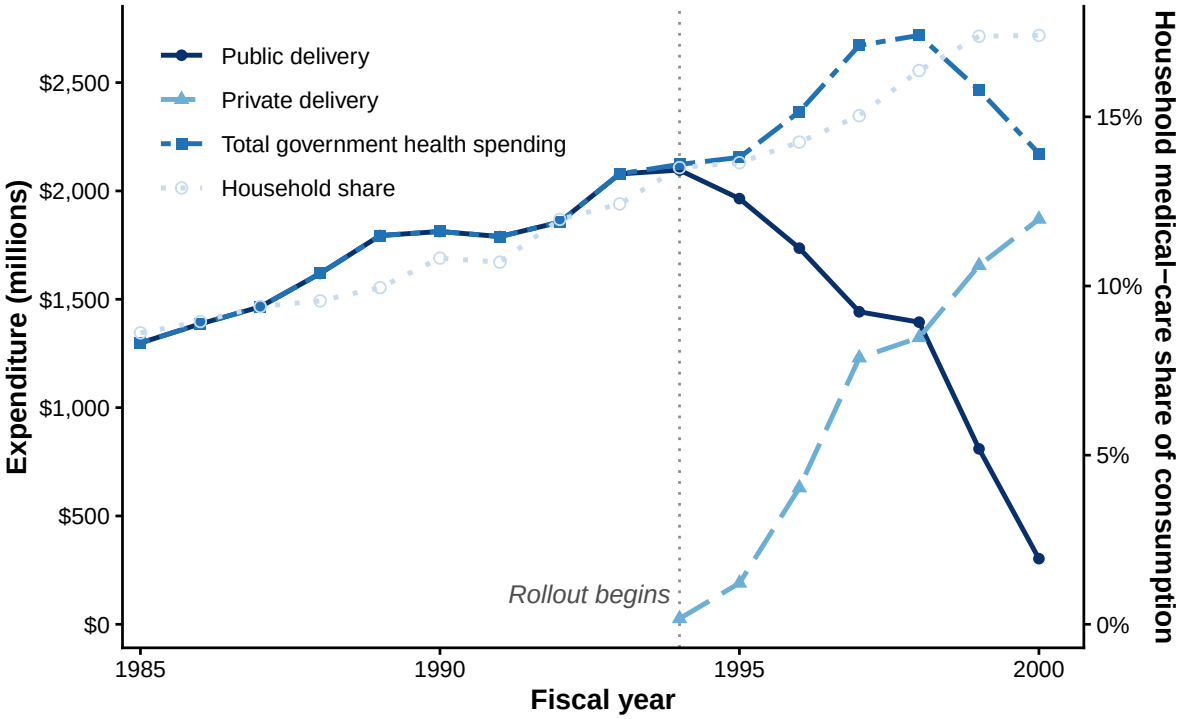


Figure 1: Delivery spending and managed-care purchasing

Dollar series are in constant 2025 dollars. The financing appendix defines each spending series and its source.

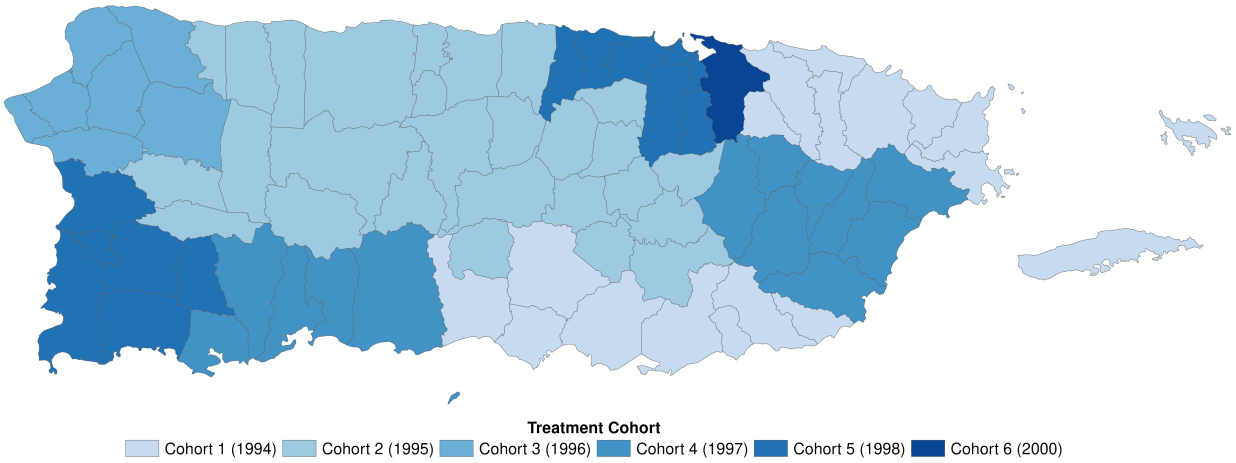
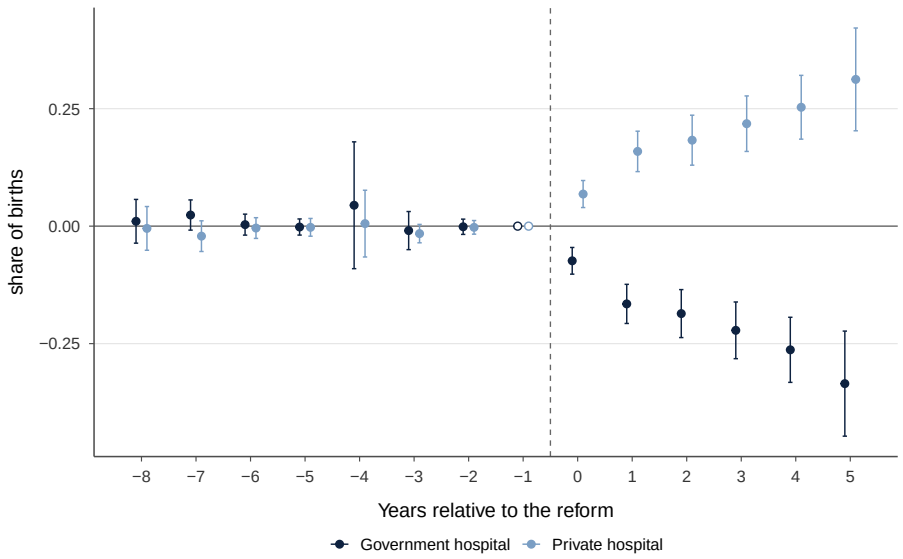
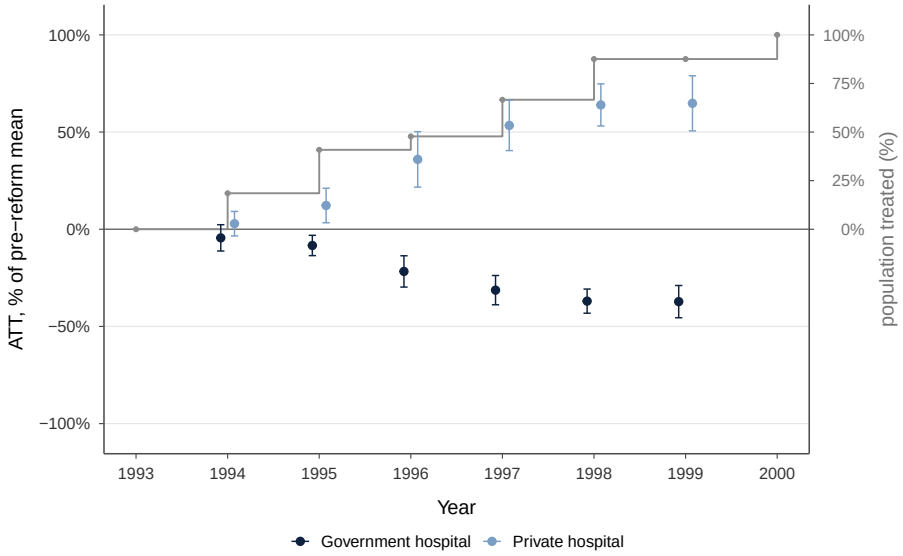


Figure 2: Adoption cohorts

Each of the 78 counties is shaded by its adoption cohort, the year the reform reached it: six annual cohorts from 1994 to 2000, the last of them San Juan alone.



(a) Rollout, by relative years



(b) Rollout, by calendar years

Figure 3: Hospital birth share by ownership

Both panels report Callaway and Sant'Anna (2021) estimates against not-yet-treated counties, Puerto Rico counties, 1990 to 1999. In panel (a) the year before adoption is the reference period and is set to zero. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.

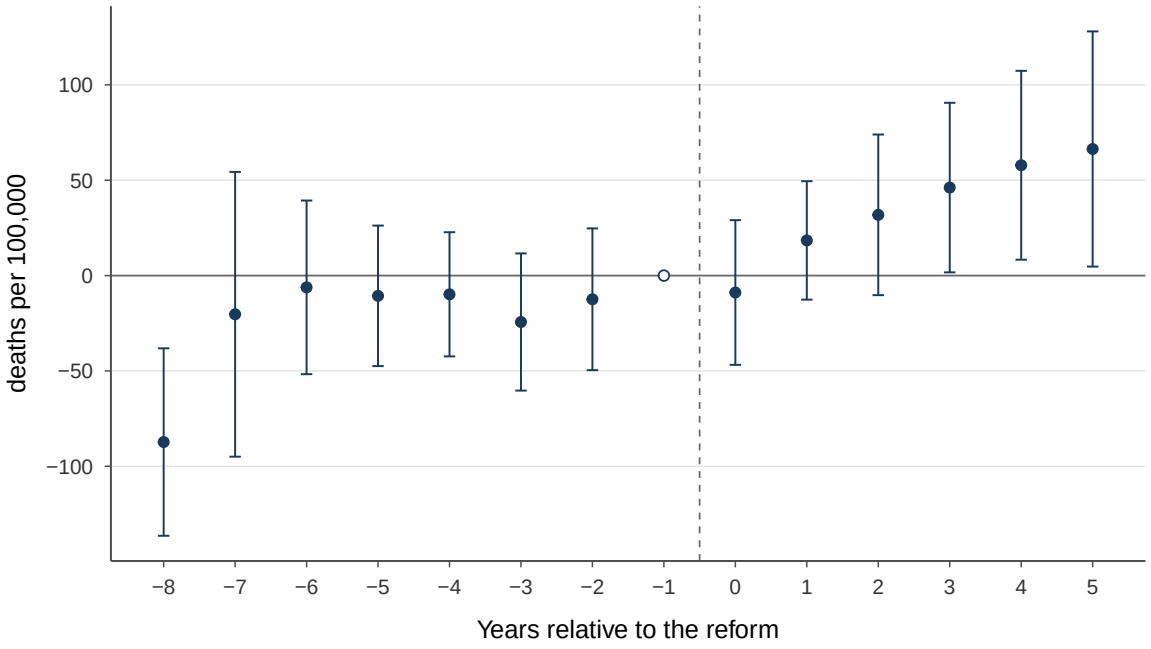
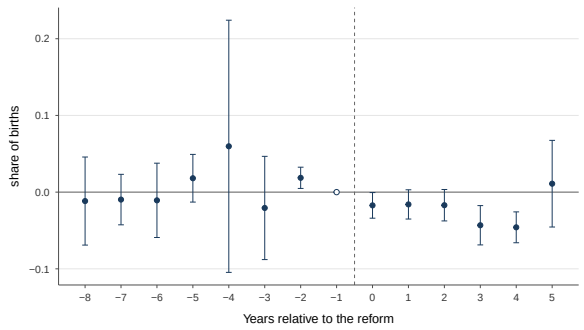
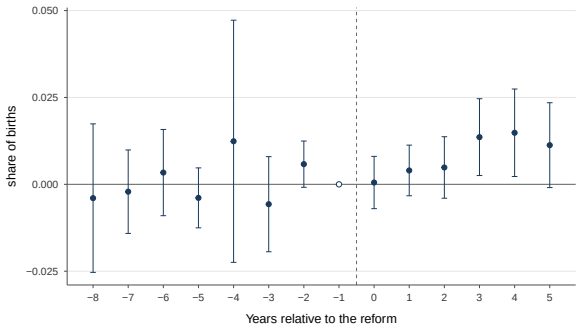


Figure 4: All-cause mortality

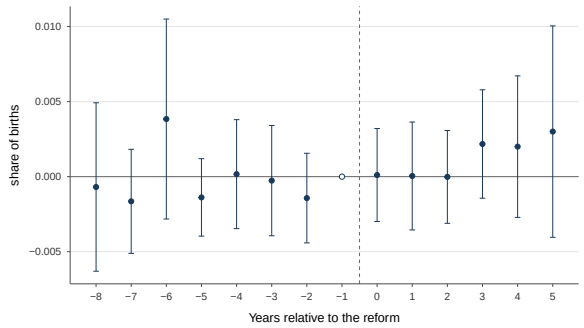
Callaway and Sant'Anna (2021) estimates by year relative to adoption, averaged over the cohorts observed at each year. The control group is the not-yet-treated counties. The year before adoption is the reference period and is set to zero. Puerto Rico counties, 1990 to 1999. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.



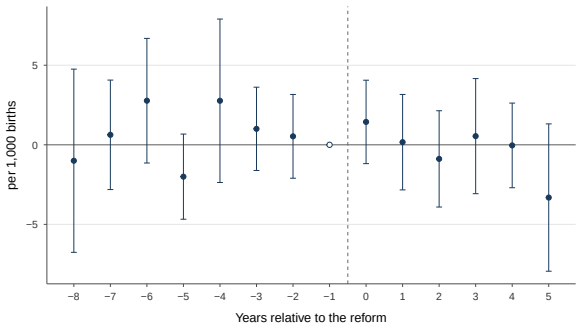
(a) Early prenatal care



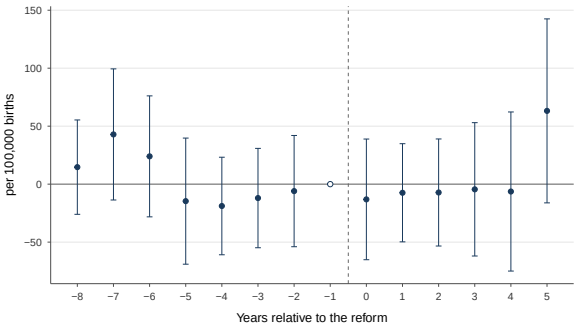
(b) Low birth weight



(c) Very low birth weight



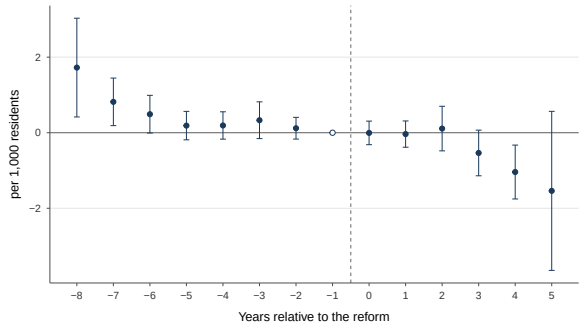
(d) Infant mortality



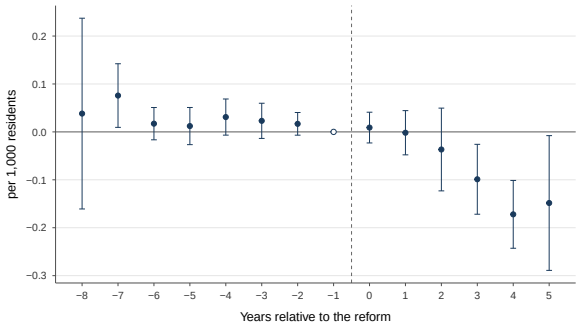
(e) Maternal mortality

Figure 5: Prenatal care and prenatal-care-sensitive outcomes

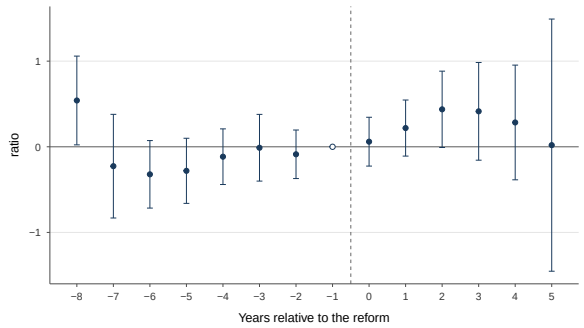
Callaway and Sant'Anna (2021) estimates by year relative to adoption, averaged over the cohorts observed at each year. The control group is the not-yet-treated counties. The year before adoption is the reference period and is set to zero. Puerto Rico counties, 1990 to 1999. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.



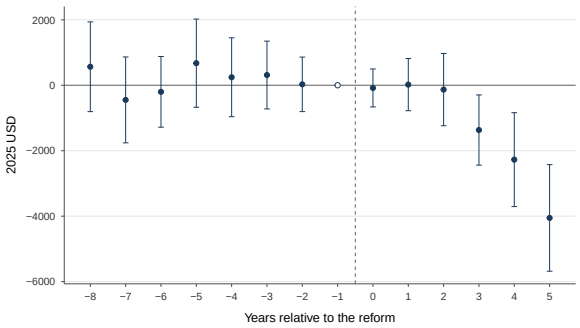
(a) Employment



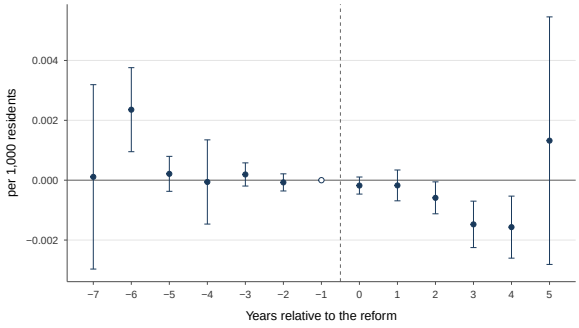
(b) Establishments



(c) Employees per establishment



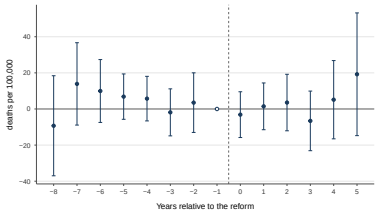
(d) Wages per worker



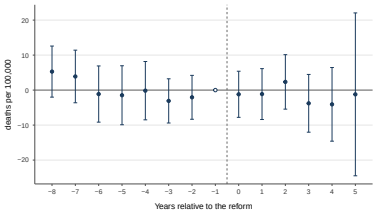
(e) Certified hospitals

Figure 6: Health supply

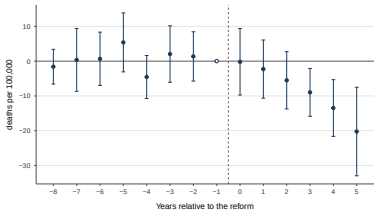
Callaway and Sant'Anna (2021) estimates by year relative to adoption, averaged over the cohorts observed at each year. The control group is the not-yet-treated counties. The year before adoption is the reference period and is set to zero. Employment, establishments, employees per establishment and wages are QCEW NAICS 621, ambulatory health care services, for the counties with published figures in every year, 1990 to 1999. Certified hospitals come from the Medicare Provider of Services file, 1991 to 1999, for every county. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.



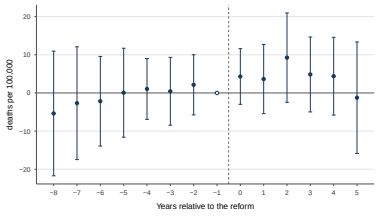
(a) Heart disease



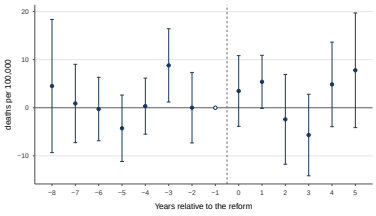
(b) Cerebrovascular disease



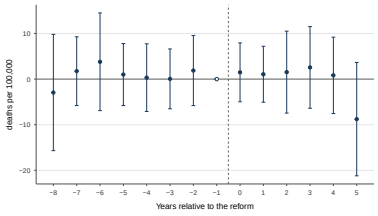
(c) Hypertension



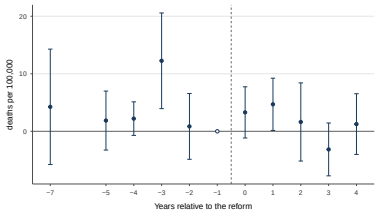
(d) Diabetes



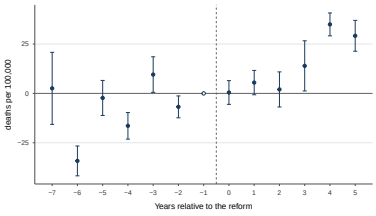
(e) Chronic pulmonary disease



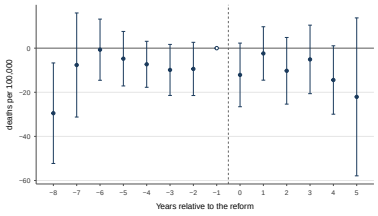
(f) Pneumonia and influenza



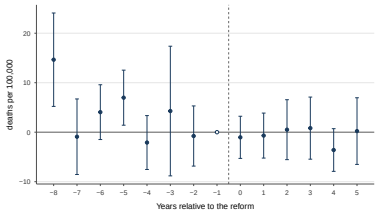
(g) Septicemia



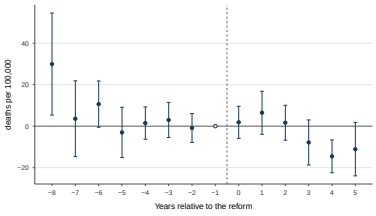
(h) AIDS



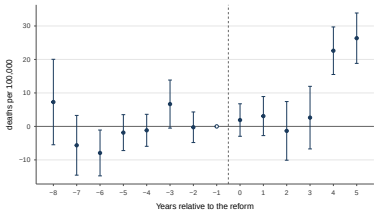
(i) Cancer



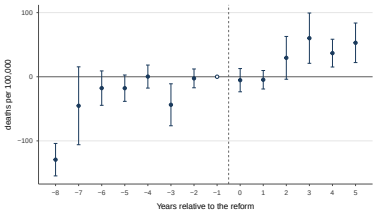
(j) Chronic liver disease



(k) Accidents



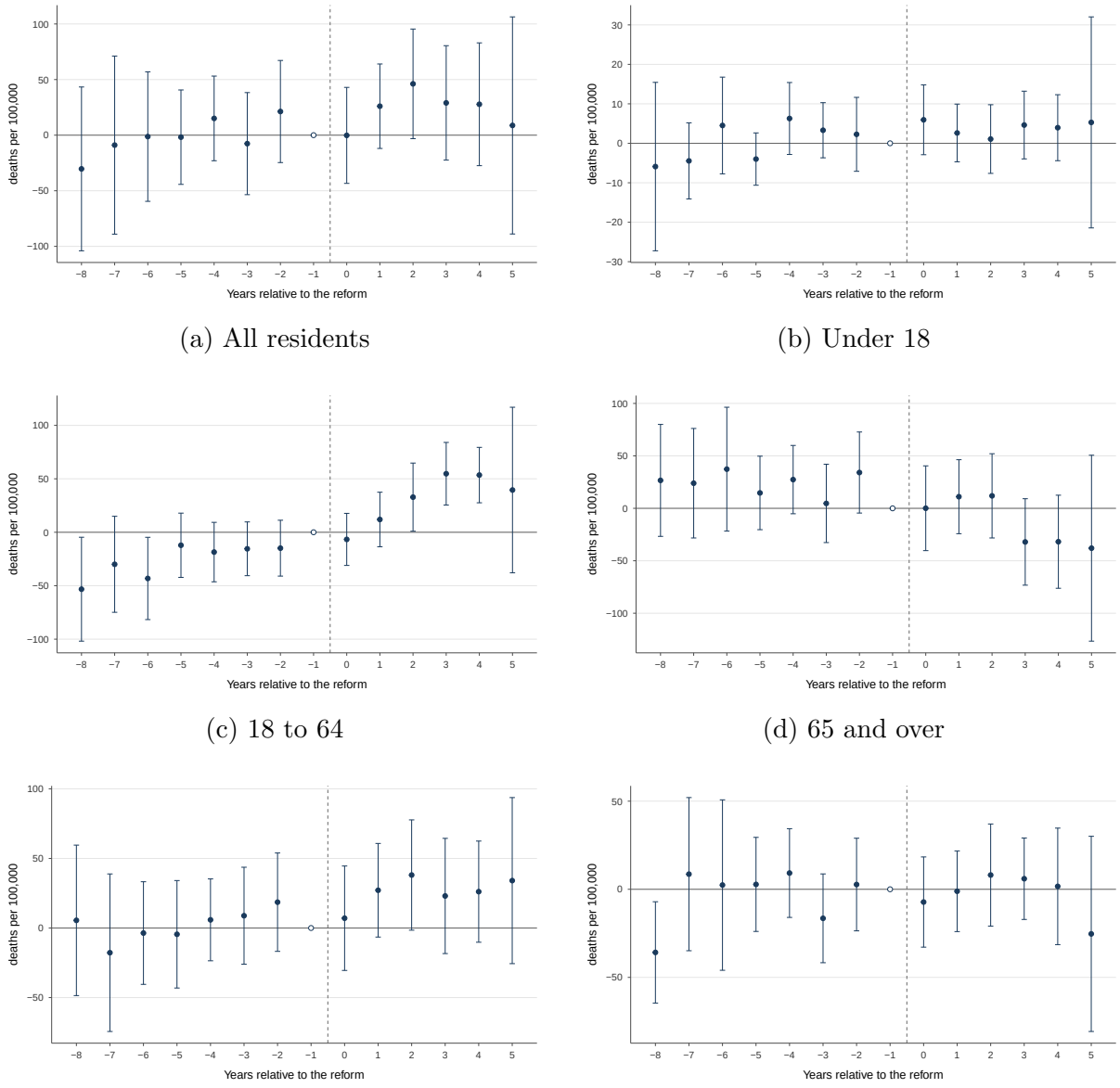
(l) Homicide



(m) Unclassified

Figure 7: Mortality by cause

Callaway and Sant'Anna (2021) estimates by year relative to adoption, averaged over the cohorts observed at each year. The control group is the not-yet-treated counties. The year before adoption is the reference period and is set to zero. Puerto Rico counties, 1990 to 1999. Unclassified is all-cause deaths less the causes in the other panels. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.



(a) All residents

(b) Under 18

(c) 18 to 64

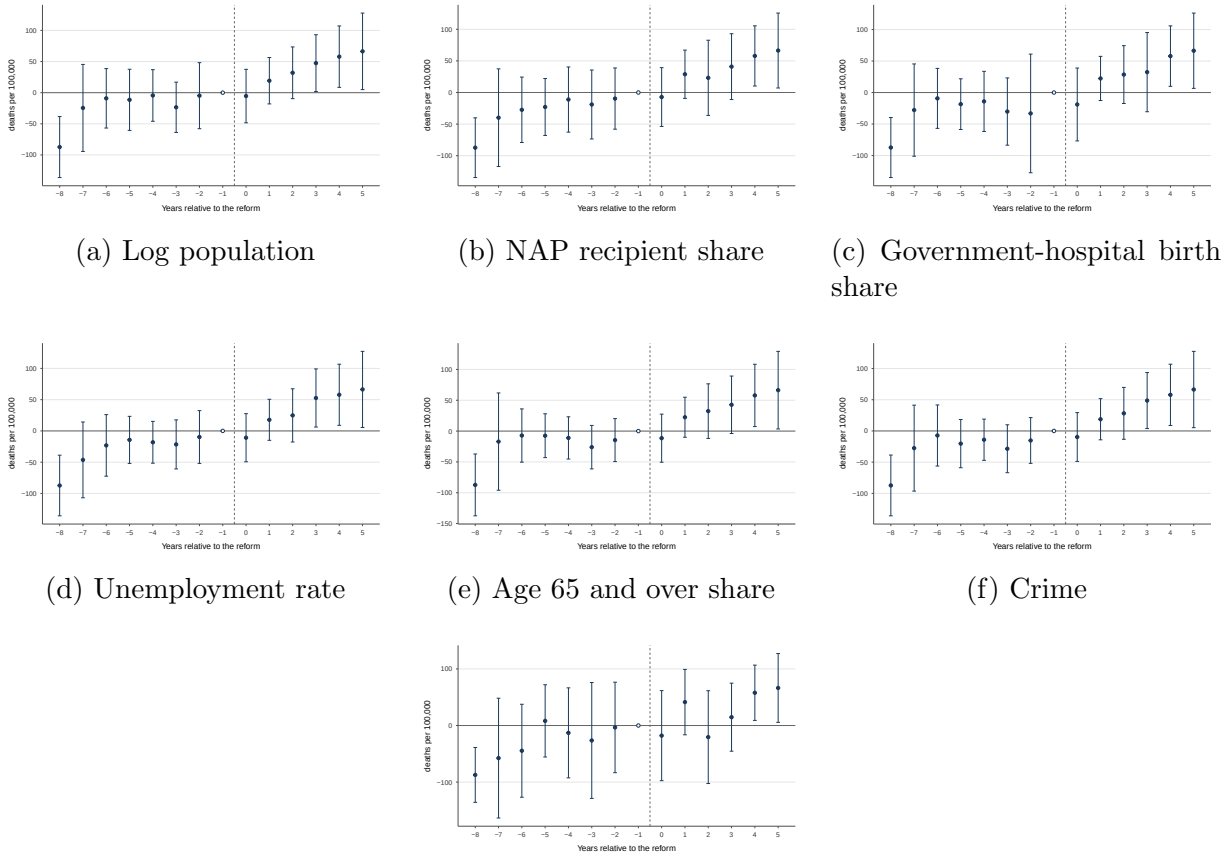
(d) 65 and over

(e) Men

(f) Women

Figure 8: All-cause mortality by age and sex

Callaway and Sant'Anna (2021) estimates by year relative to adoption, averaged over the cohorts observed at each year. The control group is the not-yet-treated counties. The year before adoption is the reference period and is set to zero. Puerto Rico counties, 1990 to 1999. Deaths are the Demographic Registry's death records of residents, each weighted by the inverse of the share of the Planning Board's 1990 to 1999 deaths the records hold, per 100,000 residents in every panel. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.



(a) Log population

(b) NAP recipient share

(c) Government-hospital birth share

(d) Unemployment rate

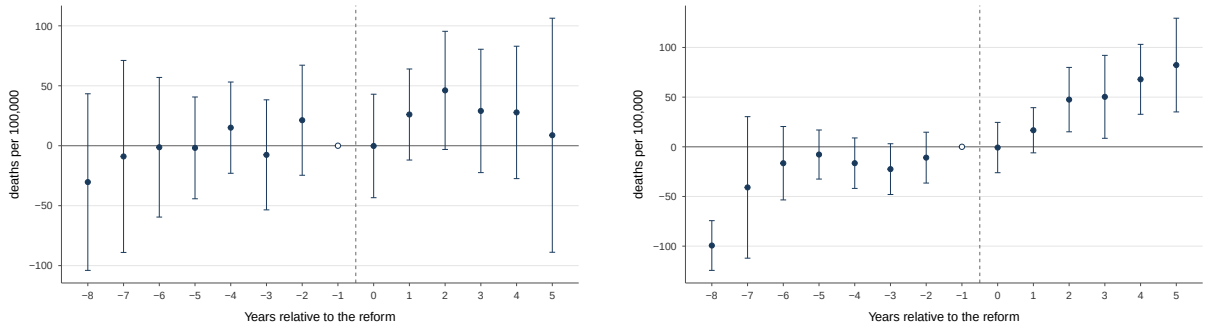
(e) Age 65 and over share

(f) Crime

(g) All six covariates

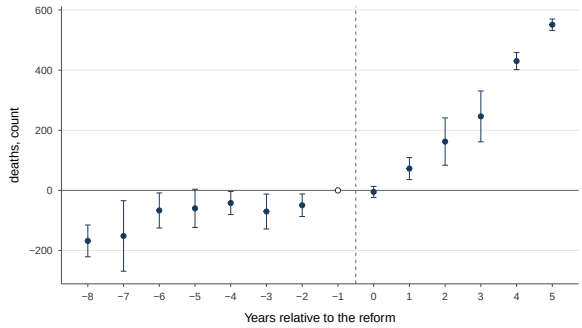
Figure 9: All-cause mortality with covariate-specific trends

Callaway and Sant'Anna (2021) doubly robust estimates by year relative to adoption, conditioning on the baseline covariates each panel names, the characteristics on which the adoption cohorts differ in Appendix Table A3: 1990 to 1993 means of log population, the NAP recipient share, the government-hospital birth share, the unemployment rate and the crime rate, and the 1990 Census share aged 65 and over. The control group is the not-yet-treated counties. The year before adoption is the reference period and is set to zero. Puerto Rico counties, 1990 to 1999. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.

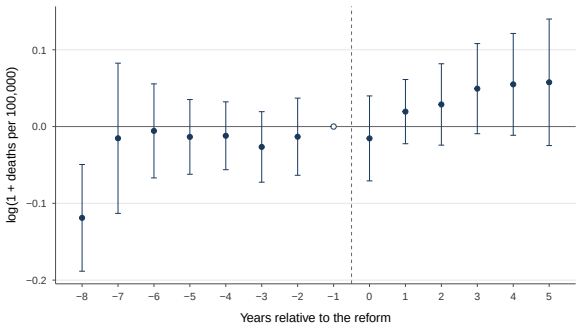


(a) Death records

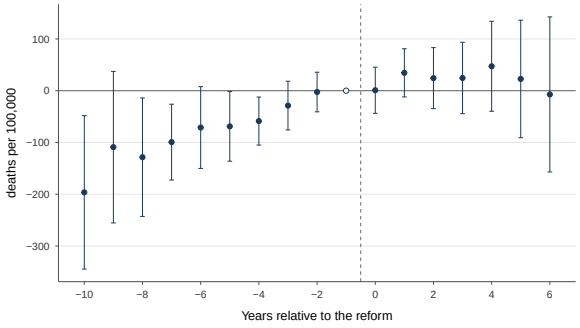
(b) Population weighted



(c) Levels



(d) Logs



(e) Never-treated, conditional

Figure 10: All-cause mortality under alternative specifications

Callaway and Sant'Anna (2021) estimates by year relative to adoption, one specification per panel. Outside the never-treated panel, the control group is the not-yet-treated Puerto Rico counties, 1990 to 1999, and the population-weighted panel weights counties by their 1990 to 1993 mean population. The death-records panel counts the Demographic Registry's death records of residents in place of the Planning Board's deaths, each weighted by the inverse of the share of the Planning Board's 1990 to 1999 deaths the records hold. The never-treated panel compares Puerto Rico counties with never-treated mainland counties in states that had not adopted Medicaid managed care by 2000, conditional on population density (1990 to 1993 mean) and the 1990 Census poverty rate, through 2000. The year before adoption is the reference period and is set to zero. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.

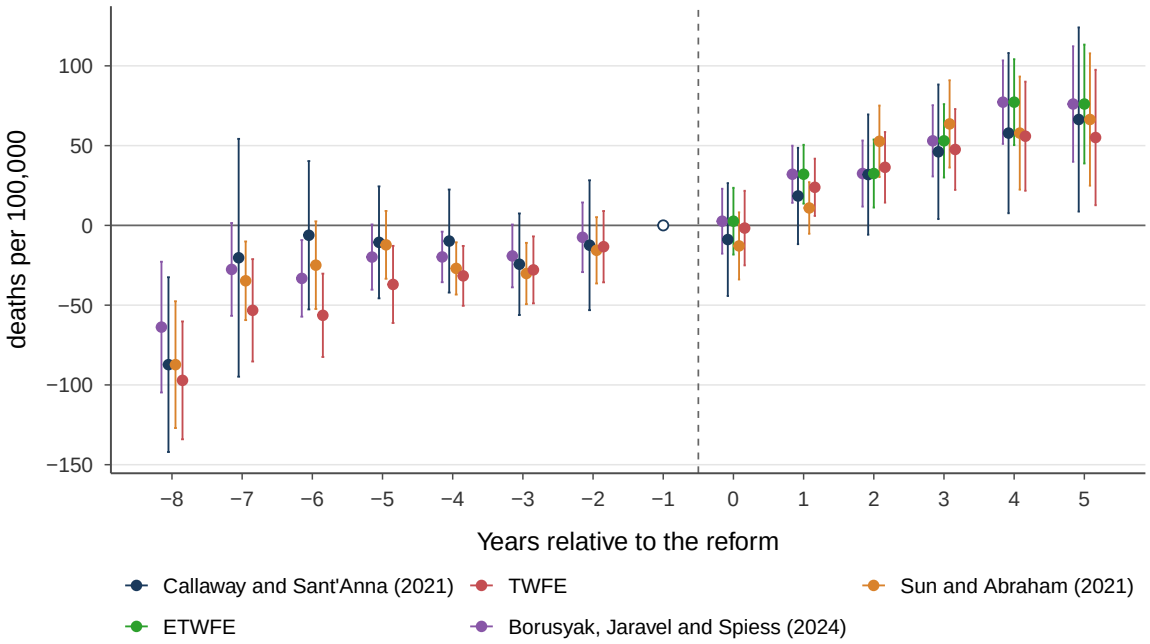


Figure 11: Alternative estimators

Event-study estimates by year relative to adoption, one series per estimator. TWFE is two-way fixed effects and ETWFE is the extended two-way fixed effects estimator of Wooldridge (2025), which estimates effects from adoption onward only. For Callaway and Sant'Anna (2021), the control group is the not-yet-treated counties and the year before adoption is the reference period, set to zero. Puerto Rico counties, 1990 to 1999. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications for Callaway and Sant'Anna (2021), and 95 percent pointwise confidence intervals from standard errors clustered on the county for the other series.

A Additional tables and figures

Tables

Table A1: Health professionals per 1,000 residents and island totals by year

Variable	1989–1992	1992–1995	1995–1998	1998–2001
Panel A: Licensed health professionals (Puerto Rico Planning Board)				
Island totals, headcount				
Physicians, total	8,443	6,764	7,144	7,623
Dentists, total	1,329	1,075	1,028	1,360
Registered nurses, total	16,848	12,541	12,145	12,694
Practical nurses, total	8,881	7,128	6,536	7,206
Medical technologists, total	2,751	2,179	2,294	2,734
Total	38,252	29,687	29,147	31,617
Health professionals				
Physicians per 1,000 residents	1.061	0.980	1.047	1.096
	(1.205)	(0.929)	(0.949)	(0.997)
Dentists per 1,000 residents	0.198	0.183	0.180	0.239
	(0.164)	(0.121)	(0.119)	(0.141)
Registered nurses per 1,000 residents	2.240	1.981	1.897	1.922
	(2.674)	(2.053)	(1.933)	(2.016)
Practical nurses per 1,000 residents	1.368	1.281	1.153	1.203
	(1.340)	(1.129)	(0.984)	(1.106)
Medical technologists per 1,000 residents	0.375	0.346	0.373	0.464
	(0.398)	(0.312)	(0.321)	(0.361)
Health professionals per 1,000 residents	5.241	4.771	4.650	4.924
	(5.597)	(4.330)	(4.115)	(4.432)
Treated counties	0	45	77	78
Not-yet-treated counties	78	33	1	0
% of population observed	100.0	100.0	100.0	100.0

Notes: County means with standard deviations in parentheses. Percent of population observed is the population of the counties this panel covers in that year, divided by the population of every county in that year. Workforce counts are licensed professionals, published for three-year windows, one window to a column; the total sums the five professions, whose licenses do not overlap.

Table A2: Ambulatory Health Care Services employment and establishments by ownership sector, Puerto Rico (NAICS 621)

Variable	1990	1992	1994	1996	1998	2000
Panel A: County Ambulatory Health Care Services (QCEW)						
Employment per 1,000 residents	2.24	2.75	3.20	3.71	4.47	4.83
	(2.15)	(2.79)	(3.02)	(2.96)	(3.59)	(3.56)
Establishments per 1,000 residents	0.670	0.765	0.782	0.887	0.990	1.089
	(0.473)	(0.515)	(0.518)	(0.576)	(0.616)	(0.657)
Treated counties	0	0	9	37	58	59
Not-yet-treated counties	59	59	50	22	1	0
% of population observed	90.7	90.7	90.6	90.6	90.5	90.5
Panel B: County Ambulatory Health Care Services by ownership (QCEW)						
Employment	209.5	242.6	285.2	317.4	374.3	407.2
	(583.0)	(629.9)	(742.9)	(770.3)	(856.3)	(901.3)
Total	12,362	14,313	16,827	18,726	22,084	24,025
Government total						
Private total	12,362	14,313	16,827	18,726	22,084	24,025
Establishments	56.86	63.44	65.42	71.36	79.32	87.34
	(149.65)	(158.73)	(157.91)	(162.51)	(174.00)	(184.59)
Total	3,355	3,743	3,860	4,210	4,680	5,153
Government total						
Private total	3,355	3,743	3,860	4,210	4,680	5,153
Treated counties	0	0	9	37	58	59
Not-yet-treated counties	59	59	50	22	1	0
% of population observed	90.7	90.7	90.6	90.6	90.5	90.5
% of employment observed	97.9	97.8	97.7	96.9	96.4	96.8
% of establishments observed	97.7	97.3	97.2	96.4	96.2	96.3
Panel C: State Ambulatory Health Care Services (QCEW)						
Employment per 1,000 residents	3.58	4.09	4.67	5.24	6.11	6.52
Employment, total	12,627	14,631	17,230	19,317	22,897	24,815
Establishments per 1,000 residents	0.973	1.075	1.078	1.185	1.298	1.404
Establishments, total	3,433	3,848	3,973	4,367	4,866	5,349
Treated counties	0	0	18	52	77	78
Not-yet-treated counties	78	78	60	26	1	0
% of population observed	100.0	100.0	100.0	100.0	100.0	100.0

Table A2: Ambulatory Health Care Services employment and establishments by ownership sector, Puerto Rico (NAICS 621) (continued)

Variable	1990	1992	1994	1996	1998	2000
Panel D: County private healthcare sector (CBP)						
Employment per 1,000 residents	12.62	11.50	12.43	14.30	16.17	18.69
	(8.79)	(7.98)	(9.38)	(9.05)	(10.00)	(11.27)
Employment	1,754.3	1,518.6	1,616.3	1,711.9	1,912.0	2,144.5
	(2,551.5)	(2,484.9)	(2,712.4)	(2,940.5)	(3,077.7)	(3,256.3)
Private total	24,560	27,335	30,710	35,949	40,153	45,035
Establishments per 1,000 residents	0.813	0.799	0.878	1.308	1.445	1.474
	(0.523)	(0.528)	(0.534)	(0.740)	(0.744)	(0.769)
Establishments	111.43	112.86	119.52	167.00	180.52	185.29
	(210.12)	(208.67)	(205.67)	(273.55)	(282.67)	(286.01)
Private total	2,340	2,370	2,510	3,507	3,791	3,891
Treated counties	0	0	4	12	20	21
Not-yet-treated counties	21	21	17	9	1	0
% of population observed	56.8	56.8	56.6	55.5	55.1	54.7
Panel E: Individual healthcare employment (Census)						
Employment per 1,000 residents	4.66					7.33
Employment, total	16,449					27,916
% of employees with no county identified	100.0					100.0
% of population observed	100.0					100.0

Notes: County means with standard deviations in parentheses. Percent of population observed is the population of the counties this panel covers in that year, divided by the population of every county in that year. Percent of employment observed is the covered counties' employment over the island total that year, and percent of establishments observed is the same quotient on establishments; health workers are employed civilians aged 16 and over in the ambulatory categories that the data-construction appendix lists. They include the self-employed. Panel D covers counties where CBP published a figure, which it does above 50 employees in 1990–1992 and 100 in 1993–2000. SIC 80 includes hospitals and nursing and personal care; the data-construction appendix lists its components.

Table A3: Pre-reform means by treatment cohort

Variable	1994	1995	1996	1997	1998	2000	F
Ambulatory health care employment per 1,000 residents	3.07	2.12	1.93	2.58	2.99	10.04	2.17
Ambulatory health care establishments per 1,000 residents	0.643	0.608	0.678	0.823	0.826	2.494	3.80
Ambulatory health care employees per establishment	4.17	3.27	2.70	2.73	3.25	4.03	1.18
Ambulatory health care wages per worker (2025 USD)	22,774	20,723	18,055	19,805	19,842	26,369	0.92
Hospitals per 1,000 residents	0.027	0.013	0.013	0.021	0.012	0.045	1.02
All-cause mortality per 100,000	725.5	678.0	711.6	745.6	730.2	1,065.7	4.61
Government-hospital birth share (percent)	74.5	67.1	59.1	65.3	57.2	60.2	3.10
Private-hospital birth share (percent)	25.3	36.3	39.7	34.4	42.6	37.0	5.24
Population	36,717	29,535	35,263	51,767	62,300	443,627	23.53
Unemployment rate (percent)	19.43	18.15	21.54	20.37	14.80	10.17	3.97
Population per square mile	1,021.2	714.1	896.3	940.0	2,268.8	9,280.9	16.53
NAP recipients per 1,000 residents	455.6	529.2	458.3	466.4	358.7	313.1	7.15
Population aged 65 and over, 1990 (percent)	9.0	8.9	9.4	9.3	9.7	13.5	2.71
Families below the poverty threshold, 1990 (percent)	58.2	66.0	64.1	61.4	51.7	44.0	6.43
Crime per 1,000 residents	23.1	19.5	20.8	22.3	27.7	77.3	7.82

Notes: Cells are county means over 1990–1993, averaged within treatment cohort. The 5 percent critical value at the median degrees of freedom is 2.34. Crime counts Type I offenses: murder, rape, robbery, aggravated assault, burglary, larceny and auto theft.

Table A4: Death records from the Demographic Registry by year, Puerto Rico counties

Variable	1990	1992	1994	1996	1998	2000
Panel A: All deaths						
Deaths in the records	13,117	13,805	14,416	15,015	15,217	14,412
Percent of the Planning Board's deaths	50.2	50.4	50.7	50.3	50.7	50.5
Deaths per 100,000 residents	702.3	731.9	735.6	764.9	773.8	721.8
	(140.7)	(118.7)	(125.2)	(127.3)	(137.5)	(122.4)
Deaths per county	168	177	185	192	195	185
	(269)	(285)	(288)	(290)	(281)	(266)
Treated counties	0	0	18	52	77	78
Not-yet-treated counties	78	78	60	26	1	0
Percent of population observed	100.0	100.0	100.0	100.0	100.0	100.0
Panel B: By age at death						
Under 18						
Deaths in the records	661	619	603	518	495	459
Percent of the year's deaths	5.0	4.5	4.2	3.4	3.3	3.2
Deaths per 100,000 residents	36.6	36.2	33.6	25.6	25.7	24.2
	(20.9)	(19.0)	(13.7)	(13.3)	(13.1)	(16.1)
18 to 64						
Deaths in the records	4,246	4,524	4,865	5,027	4,711	4,453
Percent of the year's deaths	32.4	32.8	33.7	33.5	31.0	30.9
Deaths per 100,000 residents	212.1	218.1	239.9	250.7	238.5	217.2
	(56.4)	(62.3)	(65.0)	(59.8)	(47.1)	(49.2)
65 and over						
Deaths in the records	8,202	8,648	8,934	9,459	10,003	9,498
Percent of the year's deaths	62.5	62.6	62.0	63.0	65.7	65.9
Deaths per 100,000 residents	453.5	477.3	462.0	488.5	509.3	480.3
	(114.7)	(106.8)	(105.8)	(109.7)	(111.4)	(98.7)
Panel C: By sex						
Men						
Deaths in the records	7,728	8,183	8,473	8,735	8,613	8,180
Percent of the year's deaths	58.9	59.3	58.8	58.2	56.6	56.8
Deaths per 100,000 residents	415.6	435.2	435.4	443.8	440.9	409.4
	(80.7)	(83.9)	(86.3)	(82.1)	(81.7)	(78.4)
Women						
Deaths in the records	5,388	5,622	5,942	6,280	6,604	6,232
Percent of the year's deaths	41.1	40.7	41.2	41.8	43.4	43.2
Deaths per 100,000 residents	286.6	296.7	300.2	321.2	333.0	312.5
	(84.4)	(65.5)	(64.1)	(67.0)	(72.9)	(78.7)

Notes: Deaths are the records of residents of the 78 counties in the Demographic Registry's death records, as the Puerto Rico Institute of Statistics publishes them (Instituto de Estadísticas de Puerto Rico, 2026). Deaths in the records and percents are island totals for the year. Percent of the year's deaths is a group's deaths over all deaths in the records that year, and percent of the Planning Board's deaths is the island's deaths in the records over the island's deaths in the Planning Board series. Deaths per 100,000 residents and deaths per county are county means, with standard deviations in parentheses. A rate divides a county's deaths in the records by 0.5055, the share of the Planning Board's 1990–1999 deaths the records hold, and by the county's whole population from the Planning Board's municipal series, so the groups' rates add up to the all-cause rate and their percents to 100, apart from the 106 deaths of unknown age and 2 of indeterminate sex, which the table does not show. Percent of population observed is the population of the counties covered in that year, divided by the population of every county in that year. Age at death comes from the dates of birth and death, or from the recorded age where the birth date gives only the year.

Table A5: Deaths by cause and share of all deaths by year, Puerto Rico

Variable	1990	1992	1994	1996	1998	2000
Panel A: Population, mortality and births (Puerto Rico Planning Board)						
Deaths, with share of all deaths beneath						
All causes	26,148	27,397	28,444	29,871	29,990	28,550
	(100.00%)	(100.00%)	(100.00%)	(100.00%)	(100.00%)	(100.00%)
Diabetes	1,664	1,830	1,866	2,091	2,295	2,273
	(6.36%)	(6.68%)	(6.56%)	(7.00%)	(7.65%)	(7.96%)
Cancer	4,063	4,362	4,271	4,563	4,658	4,693
	(15.54%)	(15.92%)	(15.02%)	(15.28%)	(15.53%)	(16.44%)
Heart disease	5,910	5,812	5,783	5,925	6,038	5,222
	(22.60%)	(21.21%)	(20.33%)	(19.84%)	(20.13%)	(18.29%)
Cerebrovascular	1,186	1,275	1,419	1,465	1,420	1,166
	(4.54%)	(4.65%)	(4.99%)	(4.90%)	(4.73%)	(4.08%)
Hypertension	741	900	899	1,084	1,395	1,575
	(2.83%)	(3.29%)	(3.16%)	(3.63%)	(4.65%)	(5.52%)
Cirrhosis	290	804	648	670	672	820
	(1.11%)	(2.93%)	(2.28%)	(2.24%)	(2.24%)	(2.87%)
Pneumonia and influenza	1,178	1,192	1,177	1,174	1,314	785
	(4.51%)	(4.35%)	(4.14%)	(3.93%)	(4.38%)	(2.75%)
Septicemia	0		499	634	599	601
	(0.00%)	(–)	(1.75%)	(2.12%)	(2.00%)	(2.11%)
Chronic pulmonary	875	998	1,183	1,248	1,385	1,144
	(3.35%)	(3.64%)	(4.16%)	(4.18%)	(4.62%)	(4.01%)
AIDS	0	1,407	1,530	1,474	582	659
	(0.00%)	(5.14%)	(5.38%)	(4.93%)	(1.94%)	(2.31%)
Homicide	568	838	998	928	819	700
	(2.17%)	(3.06%)	(3.51%)	(3.11%)	(2.73%)	(2.45%)
Accidents	1,503	1,171	1,287	1,520	1,387	1,321
	(5.75%)	(4.27%)	(4.52%)	(5.09%)	(4.62%)	(4.63%)
Unclassified	8,170	6,823	6,884	7,095	7,426	7,591
	(31.25%)	(24.90%)	(24.20%)	(23.75%)	(24.76%)	(26.59%)
Treated counties	0	0	18	52	77	78
Not-yet-treated counties	78	78	60	26	1	0
% of population observed	100.0	100.0	100.0	100.0	100.0	100.0

Notes: Entries are TOTAL deaths in Puerto Rico, with that cause's share of all deaths in parentheses beneath. Both are island figures, not county averages. Percent of population observed is the population of the counties this panel covers in that year, divided by the population of every county in that year. Unclassified is all-cause deaths less the causes listed, so the shares sum to 100; where a cause is unreported its deaths fall into that row; in two county-years the published cause counts exceed published all-cause deaths, by 15 in Hatillo in 1992 and by 3 in Las Marias in 1995, and the residual is floored at zero there rather than carried negative. Sample: unbalanced, every county the panel observes, with no balance restriction.

Table A6: Mortality and birth-margin outcomes by year, Puerto Rico counties

Variable	1990	1992	1994	1996	1998	2000
Panel A: Population, mortality and births (Puerto Rico Planning Board)						
Mortality						
All-cause mortality per 100,000	703.5	720.2	728.4	763.5	767.2	718.3
	(116.7)	(103.9)	(120.9)	(126.5)	(114.3)	(113.2)
Diabetes mortality per 100,000	46.9	54.3	50.7	57.0	62.6	59.5
	(23.4)	(22.0)	(19.2)	(24.1)	(23.2)	(18.9)
Cancer mortality per 100,000	108.1	116.7	111.2	118.1	120.0	117.7
	(27.6)	(25.0)	(34.6)	(29.8)	(24.1)	(23.8)
Heart disease mortality per 100,000	163.9	156.8	151.7	153.8	153.9	133.3
	(36.6)	(38.8)	(31.6)	(37.4)	(40.7)	(38.4)
Cerebrovascular mortality per 100,000	34.7	36.0	37.7	39.5	36.0	30.4
	(18.8)	(12.5)	(15.8)	(12.6)	(13.8)	(12.4)
Hypertension mortality per 100,000	21.2	26.7	25.7	30.4	36.6	40.0
	(9.3)	(11.7)	(12.2)	(14.2)	(12.1)	(13.3)
Cirrhosis mortality per 100,000	7.6	22.9	17.3	19.2	18.0	20.7
	(11.2)	(31.5)	(7.8)	(10.8)	(10.2)	(10.5)
Pneumonia and influenza mortality per 100,000	29.8	29.4	28.1	28.7	32.5	20.1
	(14.9)	(12.8)	(14.4)	(11.3)	(15.5)	(10.3)
Septicemia mortality per 100,000	0.0		12.4	16.2	14.9	14.3
	(0.0)	(–)	(7.4)	(9.1)	(7.6)	(7.2)
Chronic pulmonary mortality per 100,000	23.9	29.0	33.7	34.0	37.3	31.2
	(12.5)	(14.8)	(14.5)	(14.4)	(16.2)	(15.9)
AIDS mortality per 100,000	0.0	26.6	29.4	30.2	12.0	13.4
	(0.0)	(16.7)	(19.0)	(17.0)	(9.4)	(9.2)
Homicide mortality per 100,000	9.7	13.0	17.6	15.9	15.5	14.3
	(9.7)	(13.4)	(18.1)	(15.1)	(16.1)	(13.2)
Accident mortality per 100,000	43.3	33.7	35.4	39.4	37.6	31.7
	(22.7)	(13.7)	(15.7)	(14.2)	(15.9)	(12.3)
Unclassified mortality per 100,000	214.5	175.7	177.3	181.0	190.3	191.7
	(61.6)	(40.9)	(37.1)	(38.0)	(38.8)	(44.2)
Birth outcomes						
Infant mortality per 1,000 live births	12.75	12.39	11.55	9.91	10.48	9.91
	(6.09)	(4.85)	(5.26)	(4.90)	(5.02)	(5.60)
Low birth weight share (percent)	9.28		9.86	10.49	10.98	2.46
	(1.62)	(–)	(1.44)	(1.90)	(2.24)	(1.34)
Early prenatal care share (percent)	71.9		78.5	78.6	79.7	78.6
	(6.7)	(–)	(7.2)	(5.9)	(6.5)	(8.7)
Treated counties	0	0	18	52	77	78
Not-yet-treated counties	78	78	60	26	1	0
% of population observed	100.0	100.0	100.0	100.0	100.0	100.0

Notes: County means with standard deviations in parentheses. Percent of population observed is the population of the counties this panel covers in that year, divided by the population of every county in that year. Unclassified mortality is all-cause deaths less every cause identified, expressed per 100,000 residents; in two county-years the published cause counts exceed published all-cause deaths, by 15 in Hatillo in 1992 and by 3 in Las Marias in 1995, and the residual is floored at zero there rather than carried negative.

Table A7: Two-way fixed effects estimates by cause of death

	Estimate	% of pre-treatment mean	Pre-treatment mean
Panel A. All-cause mortality			
All causes	5.277	0.7%	724.2
	(9.498)		
Panel B. Cardiovascular and metabolic causes			
Heart disease	-1.181	-0.8%	156.6
	(3.460)		
Cerebrovascular	0.1958	0.5%	36.72
	(1.556)		
Hypertension	0.9735	3.6%	26.68
	(2.399)		
Diabetes	2.588	5.0%	51.55
	(2.497)		
Panel C. Respiratory and infectious causes			
Chronic pulmonary[a]	3.768**	12.9%	29.12
	(1.782)		
Pneumonia and influenza	0.6248	2.1%	29.44
	(1.306)		
Septicemia[a]	1.608	14.2%	11.31
	(1.269)		
AIDS[a]	0.5035	2.1%	24.39
	(1.887)		
Panel D. Cancer and chronic liver disease			
Cancer	-2.121	-1.9%	114.4
	(3.813)		
Cirrhosis	-1.112	-6.0%	18.64
	(1.145)		
Panel E. External causes			
Accidents	3.499	9.7%	36.02
	(2.402)		
Homicide[a]	1.417	10.3%	13.78
	(1.588)		
Panel F. Unclassified			
Unclassified	1.811	0.7%	254.2
	(6.102)		

Notes: Estimates are two-way fixed effects coefficients, by ordinary least squares, on an indicator equal to one from a county's reform year onward, with county and year fixed effects. All regressions in this table use log population, unemployment rate, crime and NAP recipient share as control variables. Callaway and Sant'Anna is the baseline identification strategy. Standard errors in parentheses: clustered by county. Puerto Rico counties, 1990 to 1999. Unmarked rows are on the balanced panel, every county in every year, and rest on the same county-years. [a] On the unbalanced panel, each on the years it is recorded. Chronic pulmonary and homicide are unavailable for 1995; septicemia is unavailable for 1990, 1991, 1992, 1993 and 1995; AIDS is unavailable for 1990, 1991 and 1995. Each covers all 78 counties in the years it appears. Unclassified is all-cause deaths less the eight unmarked causes, so it closes the decomposition, and it holds the deaths from chronic pulmonary, septicemia, AIDS and homicide. Outcomes are per 100,000. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A8: Alternative specifications: Mortality by cause and prenatal-care-sensitive outcomes

	(1) Baseline	(2) Population weighted	(3) Levels	(4) Logs	(5) Never-treated
Panel A. Mortality					
All-cause mortality					
All causes	26.96***	32.27***	173.2***	0.0247*	23.15
	(10.32)	(8.408)	(9.233)	(0.0145)	(20.54)
Pre-treatment mean	724.2	724.2	319.3	6.576	731.8
% of pre-treatment mean	3.7%	4.5%	54.2%	2.5%	3.2%
Cardiovascular and metabolic causes					
Heart disease	1.009	1.703	38.00***	-0.0211	36.37***
	(3.980)	(3.725)	(2.145)	(0.0327)	(13.09)
Pre-treatment mean	156.6	156.6	67.35	5.036	157.7
% of pre-treatment mean	0.6%	1.1%	56.4%	-2.1%	23.1%
Cerebrovascular	-1.285	-1.962	3.025***	0.0038	5.481
	(2.306)	(1.653)	(1.025)	(0.0700)	(9.859)
Pre-treatment mean	36.72	36.72	15.75	3.552	36.84
% of pre-treatment mean	-3.5%	-5.3%	19.2%	0.4%	14.9%
Hypertension	-5.943***	-6.500***	-46.11***	-0.0707	
	(2.221)	(1.231)	(1.329)	(0.0859)	
Pre-treatment mean	26.68	26.68	11.19	3.177	
% of pre-treatment mean	-22.3%	-24.4%	-412.2%	-6.8%	
Diabetes	4.913**	4.444**	-2.263	0.1079**	-10.45
	(2.409)	(1.803)	(1.469)	(0.0479)	(9.981)
Pre-treatment mean	51.55	51.55	22.05	3.858	51.73
% of pre-treatment mean	9.5%	8.6%	-10.3%	11.4%	-20.2%
Respiratory and infectious causes					
Pneumonia and influenza	0.9216	1.645	6.417***	0.0068	
	(2.363)	(1.770)	(1.658)	(0.1143)	
Pre-treatment mean	29.44	29.44	13.41	3.296	
% of pre-treatment mean	3.1%	5.6%	47.8%	0.7%	

Table A8: Alternative specifications: Mortality by cause and prenatal-care-sensitive outcomes (continued)

	(1) Baseline	(2) Population weighted	(3) Levels	(4) Logs	(5) Never-treated
Cancer and chronic liver disease					
Cancer	-9.277**	-5.490**	3.878*	-0.0874*	20.16***
	(4.245)	(2.792)	(2.127)	(0.0507)	(5.241)
Pre-treatment mean	114.4	114.4	50.77	4.719	115.3
% of pre-treatment mean	-8.1%	-4.8%	7.6%	-8.4%	17.5%
Cirrhosis	-0.6440	-0.7293	-4.479***	0.0043	107.2***
	(1.245)	(0.9615)	(0.8057)	(0.1131)	(14.10)
Pre-treatment mean	18.64	18.64	8.050	2.585	18.68
% of pre-treatment mean	-3.5%	-3.9%	-55.6%	0.4%	573.9%
External causes					
Accidents	-1.533	-2.554	-4.770	-0.0903	6.410
	(2.433)	(2.800)	(4.360)	(0.0752)	(4.488)
Pre-treatment mean	36.02	36.02	15.26	3.520	36.27
% of pre-treatment mean	-4.3%	-7.1%	-31.3%	-8.6%	17.7%
Unclassified					
Unclassified	38.79***	41.72***	179.5***	0.0754**	
	(6.722)	(7.432)	(11.48)	(0.0354)	
Pre-treatment mean	254.2	254.2	115.5	5.517	
% of pre-treatment mean	15.3%	16.4%	155.4%	7.8%	
Treated counties	78	78	78	78	78
Never-treated counties					147
Panel B. Prenatal-care-sensitive outcomes					
Infant mortality	0.0985	-0.5230	6.531***	0.0170	
	(0.7941)	(0.6708)	(1.204)	(0.1013)	
Pre-treatment mean	12.56	12.56	9.773	2.489	
% of pre-treatment mean	0.8%	-4.2%	66.8%	1.7%	
Maternal mortality	-4.291	8.345	0.7297***	1.362***	
	(22.43)	(7.469)	(0.0773)	(0.2559)	
Pre-treatment mean	19.27	19.27	0.1261	0.5629	
% of pre-treatment mean	-22.3%	43.3%	578.5%	290.4%	
Treated counties	78	78	78	78	
Never-treated counties					

Notes: Average treatment effects on the treated are estimated with Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. The never-treated column is conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census), the same covariates as the never-treated tables. Unclassified is all-cause deaths less the causes printed above it, so it closes the decomposition rather than naming a further cause. It also holds chronic pulmonary, septicemia, AIDS and homicide, which this table does not report separately. Non-MMC states had not adopted Medicaid managed care by the horizon. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10). The mortality outcomes are measured per 100,000; infant mortality is measured per 1,000 births; maternal mortality is measured per 100,000 births. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A9: Never-treated control groups with covariate-specific trends

	(1)	(2)	(3)	(4)
	Through 2000		Through 2003	
	Non-MMC	Non-HMO	Non-MMC	Non-HMO
All-cause mortality				
ATT	23.15	21.57	16.22	14.93
	(20.54)	(21.67)	(19.39)	(27.23)
Pre-treatment mean	731.8	731.8	731.8	731.8
ATT as % of pre-treatment mean	3.2%	2.9%	2.2%	2.0%
Treated counties	78	78	78	78
Never-treated counties	282	353	282	353

Notes: Average treatment effects on the treated are estimated with Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. The outcome is measured per 100,000. Puerto Rico against never-treated mainland counties, through 2000 in columns (1) and (2) and through 2003 in columns (3) and (4). Non-MMC states had not adopted Medicaid managed care by the horizon; non-HMO states operated no Medicaid HMO by the horizon. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10). Non-HMO: 378 counties in 7 states, Arkansas (75), Alabama (67), South Dakota (66), Louisiana (64), Idaho (44), Alaska (39), Wyoming (23). Never-treated mortality comes from CDC WONDER. All-cause deaths do not depend on cause coding, so the ICD-9 to ICD-10 revision does not break this series. Conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census). The island poverty rate is families below the Census threshold over all families; the mainland rate is persons for whom poverty status is determined. Estimated by outcome regression: the doubly-robust fit did not converge on these cells. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A10: Two-way fixed effects with control variables

	(1)	(2)	(3)	(4)	(5)	(6)
All-cause mortality						
Contemporaneous covariates, in levels						
ATT (per 100,000)	-1.283	5.618	-1.549	-1.756	3.619	5.277
	(10.20)	(9.899)	(10.15)	(9.907)	(10.44)	(9.498)
Pre-treatment mean	724.2	724.2	724.2	724.2	724.2	724.2
% of pre-treatment mean	-0.2%	0.8%	-0.2%	-0.2%	0.5%	0.7%
Counties	78	78	78	78	78	78
Covariates						
Log population	No	Yes	No	No	No	Yes
Unemployment rate	No	No	Yes	No	No	Yes
Crime	No	No	No	Yes	No	Yes
NAP recipient share	No	No	No	No	Yes	Yes

Notes: Estimates are two-way fixed effects coefficients, by ordinary least squares. Each covariate enters in levels at its value in that year. Standard errors in parentheses: clustered by county. Puerto Rico counties, 1990 to 1999. Each covariate is measured in the same year as the outcome, so in a treated county it is measured after that county's reform year. Each control variable is one number per county and year. Annual county values: log population, unemployment rate, crime and NAP recipient share. The columns differ in what they condition on and rest on the same counties and years. Two-way fixed effects under staggered adoption weights 32.3% of treated county-years negatively; the negative weights are 16.3% of the positive ones in absolute value. Log population is the natural log of the county's population in that year; unemployment is the county annual rate in that year; crime is Type I offenses per 1,000 residents in that year, murders included, on the definition the baseline crime control also uses; the NAP recipient share is residents receiving Nutrition Assistance Program benefits over county population in that year. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figures

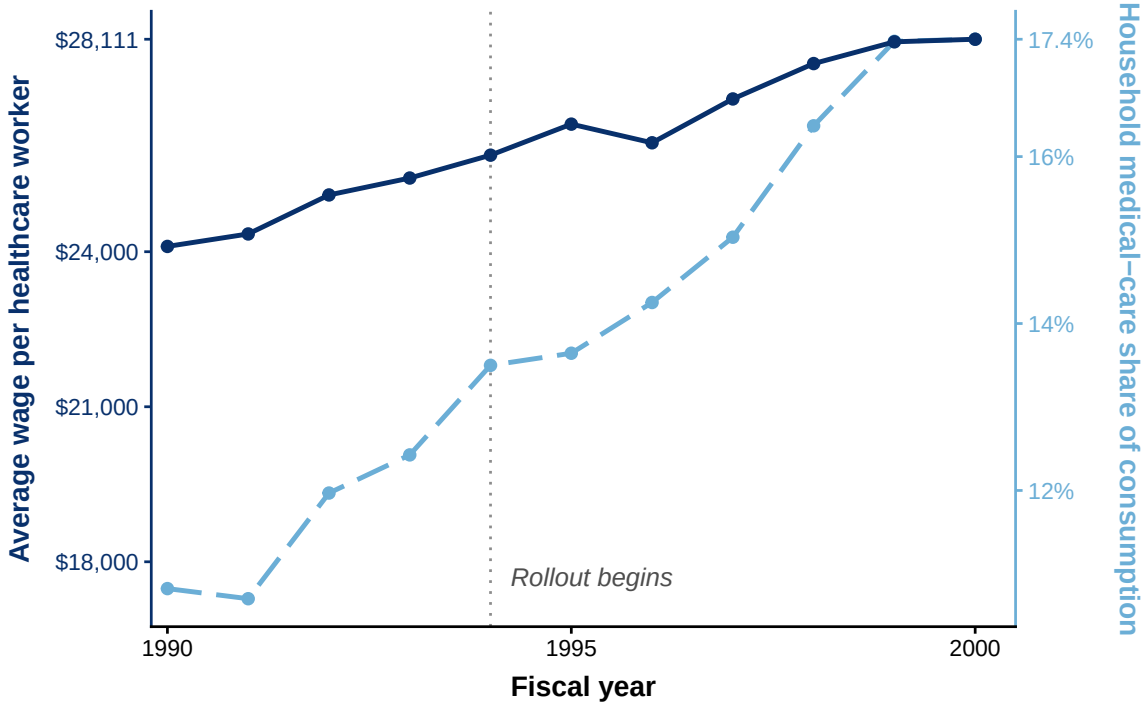


Figure A1: Household medical spending and real wages

Dollar series are in constant 2025 dollars. The wage is the QCEW NAICS 621 wage per worker, by calendar year. The financing appendix defines each series and its source.

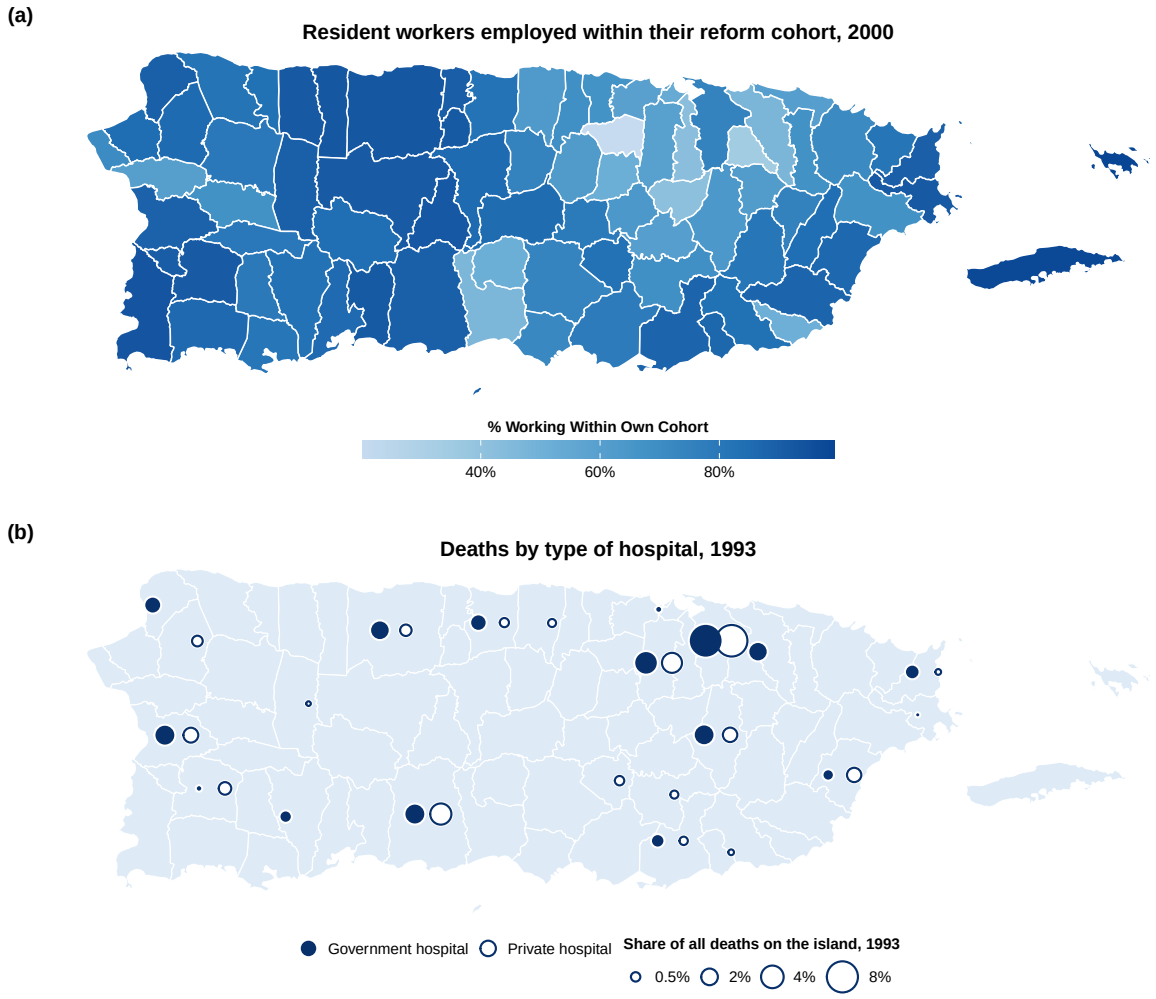
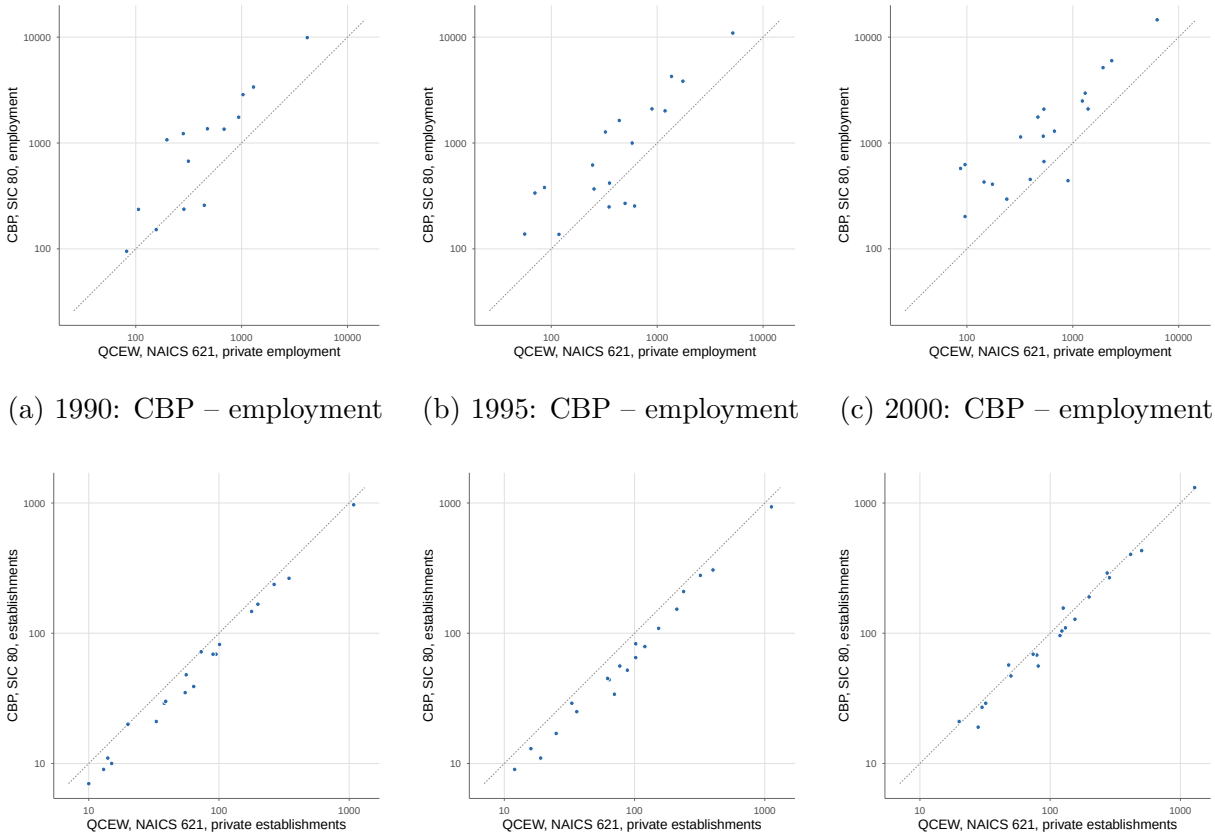


Figure A2: Commuting within reform cohorts, 2000, and deaths by type of hospital, 1993

The commuting map shades each county by the share of its resident workers who worked in a county of the same reform cohort, their own county included. Across the island in 2000, 52.5 percent of resident workers worked in their own county, 18.1 percent in another county of the same cohort and 29.3 percent in a county of a different cohort. The hospital map places a circle on each county whose hospitals recorded deaths in 1993, filled for government hospitals and open for private ones, sized by the share of all 28,494 deaths on the island that occurred there; together, government hospitals recorded 30.5 percent of the island's deaths and private hospitals 24.0 percent. Every cohort had at least one county with each kind of hospital: of the 18 counties of the 1994 cohort, 4 had a government hospital and 3 a private one; of the 27 of the 1995 cohort, 2 and 6; of the 7 of the 1996 cohort, 1 and 1; of the 13 of the 1997 cohort, 4 and 3; of the 12 of the 1998 cohort, 4 and 3; and San Juan, the 2000 cohort, had both. Sources: U.S. Census Bureau, Census 2000 county-to-county worker flows, summed from minor civil divisions; Puerto Rico Department of Health, Annual Vital Statistics Report 1993, Table 10.

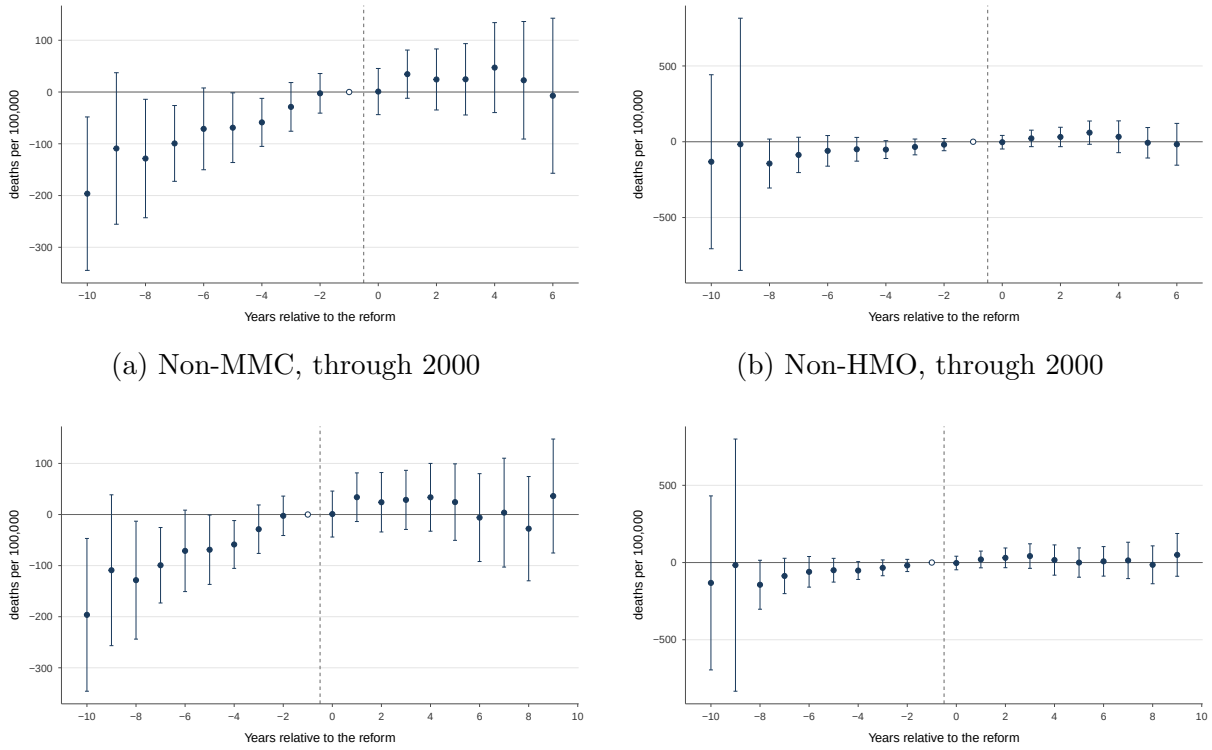


(a) 1990: CBP – employment (b) 1995: CBP – employment (c) 2000: CBP – employment

(d) 1990: CBP – establishments (e) 1995: CBP – establishments (f) 2000: CBP – establishments

Figure A3: QCEW compared to CBP

Each point is one county in the year the panel names, over the counties both sources publish a positive figure for. QCEW is NAICS 621, ambulatory health care services, restricted to private ownership; CBP is SIC 80, all health services, so it also carries hospitals and nursing and residential care. CBP counts a single reference period and QCEW an annual average. Counties drawn: employment 14 to 20 per year, establishments 20.



(a) Non-MMC, through 2000

(b) Non-HMO, through 2000

(c) Non-MMC, through 2003

(d) Non-HMO, through 2003

Figure A4: Never-treated comparison, all-cause mortality

Callaway and Sant'Anna (2021) estimates of all-cause mortality by year relative to adoption, comparing Puerto Rico counties with never-treated mainland counties, conditional on population density (1990 to 1993 mean) and the 1990 Census poverty rate. Top row through 2000, bottom row through 2003. Left, counties in states that had not adopted Medicaid managed care by the end of the window (non-MMC); right, counties in states that operated no Medicaid HMO (non-HMO). Mainland deaths come from CDC WONDER; all-cause deaths do not depend on cause coding. The year before adoption is the reference period and is set to zero. Bars are 95 percent simultaneous confidence bands from a county-clustered multiplier bootstrap with 1,000 replications.

B Reform financing and the spending series

This appendix defines the spending series in Figures 1 and A1, distinguishing the financing of the government health plan from delivery costs, and lists the plan's covered services. The island-level trends describe the reform period but do not identify its causal effects.

The reform redirected public financing toward premiums paid through the Health Insurance Administration to private insurers. Government funding continued for retained facilities and other public functions. Fiscal year 2002–2003 financing of the government health plan totaled about $2.24 billion in constant 2025 dollars: roughly 73 percent from the Commonwealth General Fund, 15 percent from federal Medicaid, 11 percent from counties, and 2 percent from CHIP (Pan American Health Organization, 2007). Operating-budget volumes yield separate estimates of $2.27 billion in fiscal 2002 and $2.18 billion in fiscal 2003 (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 2004). Federal Medicaid funding was subject to a territorial cap and matching rules that differed from the income-based matching formula for states (Medicaid and CHIP Payment and Access Commission, 2020; Pillai et al., 2024).

Between fiscal years 1996 and 1999, the Puerto Rico General Fund support for health delivery and regulation, excluding the Health Insurance Administration, fell from about $985 million to $572 million. Own-source revenue rose from about $382 million to $483 million, while federal funding remained near $650 million. Budgeted positions fell from 26,905 to 18,062, or 33 percent (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1998).

The public-delivery series covers spending by government-operated agencies, the health services assistance fund, the Medical Center, and the Cardiovascular Center. It comes from financial statements and falls below the operating-budget totals reported elsewhere. The private-delivery series records government premiums paid to managed-care organizations, excluding household payments and privately financed insurance. Household medical spend-

ing is instead a national-accounts share of personal consumption. That share rose from about 5.6 percent in 1971 to 12.4 percent in 1993 and 17.4 percent by fiscal year 2000 (Junta de Planificación de Puerto Rico, 2024).

The government health plan's benefit package covered inpatient and outpatient hospital care, laboratory and x-ray services, vision care, immunization, family planning, transportation, physician and ambulatory services, pharmacy, dental care, mental-health care, and substance-abuse treatment, without pre-existing-condition exclusions (Centers for Medicare and Medicaid Services, 2000).

C Framework derivation

The model gives the condition under which moving control of care to private managers raises mortality (Proposition 1) and shows why a shortfall in inputs cannot by itself be read as a gain in efficiency. It follows the comparison of public and private control in Hart et al. (1997), whose payoffs and ranking of efforts I state without re-deriving them. The model adds survival weights that turn efforts into mortality, separate terms for the value of quality to patients and its production cost, and inputs that quality improvement can either require or save.

Mapping to the reform. I treat the MCO and its providers as one private manager, to whom residual operational control passed through a sale, a lease, or an operating agreement. Treating them as one assumes that the MCO passes the incentive of its capitated payment through to its providers, which plans did in part by placing financial risk on primary-care physicians. Each county's residents enrolled in the one plan serving their county (Appendix E), so the model gives patients no way to leave a manager whose care deteriorates. The model holds eligibility, cost sharing, and covered benefits fixed. The reform changed eligibility and introduced co-payments, but it left a small share of residents

uninsured, 7 percent in 2003 against none in 1990 (Table 2). The household share of consumption spent on medical care kept its pre-reform trend (Figure A1), which suggests that cost sharing changed patients' spending little, and the government defined the benefits every plan covered (Appendix E). The estimates still measure the joint effect of the reform's components, which the design does not separate (Section 4).

Innovations. For each service s, the manager chooses a cost-reducing effort $e_s \geq 0$ and a quality-improving effort $i_s \geq 0$ at an effort cost of $e_s + i_s$. Cost-reducing effort yields a cost cut that saves production costs $c_s(e_s)$, with $c_s' > 0$ and $c_s'' < 0$, and harms patients by $b_s(e_s)$, with $b_s' > 0$ and $b_s'' \geq 0$, for example by reducing access or treatment intensity on labor-intensive dimensions of care that are hard to verify. In the cross-sector comparison of Fabre and Straub (2023), health and education are the two sectors whose service quality is hard to verify. Quality-improving effort yields a quality improvement that raises the value of care to patients by $v_s(i_s)$, with $v_s' > 0$, at a production cost $k_s(i_s)$, and its net benefit $\beta_s(i_s) = v_s(i_s) - k_s(i_s)$ satisfies $\beta_s' > 0$ and $\beta_s'' < 0$. Mortality depends on v_s rather than on β_s, which is why the two are kept apart. All terms vanish at zero effort, and $c_s' - b_s' \geq 0$ over the relevant range, which rules out cost cuts whose marginal harm to patients exceeds their marginal saving. The results below hold without it.

Timing. The model has four stages. First, the government and the manager sign a contract that fixes the manager's pay. Second, the manager chooses its non-contractible efforts. Third, it carries out its innovations, bargaining with the government over those that need the government's agreement, and the two split the surplus at stake equally. Fourth, it delivers care and collects its payoff. Hart et al. (1997) combine the last two stages in one date.

Payoffs. A private manager is paid a fixed amount per enrollee and bears the cost of care. Because it holds residual operational control, it carries out cost cuts without the government's agreement and keeps their savings, while their harm falls on patients. A quality improvement

raises the private manager's costs while its payment stays fixed, so the manager is paid for it only by renegotiating its contract, and it receives half of the improvement's net benefit. The government cannot obtain any part of a private manager's innovation without it, so the private manager bargains as a public manager would if it could not be replaced (Hart et al., 1997). A public manager is paid a salary while the government bears the cost of care and keeps residual operational control, so the savings from a cost cut go to the government and the manager needs its agreement for both innovations. Because it represents patients, the government values a cost cut at its savings minus its harm to them. By having a replacement carry out an innovation, the government can obtain a fraction $1-\lambda$ of its value. The remaining fraction $\lambda \in (0,1]$ is the part a replacement could not deliver, and equal bargaining over it leaves the public manager $\lambda/2$ of the innovation's value. A manager is easy to replace when λ is small. Choosing efforts at the second stage, the two managers therefore solve

$$\text{Private control:} \quad \max_{\{e_s,i_s\}} \sum_s \left[c_s(e_s) + \tfrac{1}{2}\beta_s(i_s) - e_s - i_s\right], \tag{C.1}$$

$$\text{Public control:} \quad \max_{\{e_s,i_s\}} \sum_s \left[\tfrac{\lambda}{2}\left(c_s(e_s) - b_s(e_s)\right) + \tfrac{\lambda}{2}\beta_s(i_s) - e_s - i_s\right]. \tag{C.2}$$

A cost cut whose harm equals its savings adds nothing to the public manager's objective and its full saving to the private manager's.

Equilibrium efforts. The first best maximizes the surplus from both innovations net of effort costs, $\sum_s [c_s(e_s) - b_s(e_s) + \beta_s(i_s) - e_s - i_s]$. The sign assumptions make each objective strictly concave, and with c_s' and β_s' unbounded at zero effort and below 1 at high effort, and $b_s'(0)$ finite, each has a unique interior maximum, given by the first-order conditions for the

first best ($*$), private control (P), and public control (G):

$$\begin{aligned} c_s'(e_s^*) - b_s'(e_s^*) &= 1, & \beta_s'(i_s^*) &= 1, \\ c_s'(e_s^P) &= 1, & \tfrac{1}{2}\beta_s'(i_s^P) &= 1, \\ \tfrac{\lambda}{2}\left[c_s'(e_s^G) - b_s'(e_s^G)\right] &= 1, & \tfrac{\lambda}{2}\beta_s'(i_s^G) &= 1. \end{aligned} \tag{C.3}$$

These conditions give the ranking of Hart et al. (1997), $e_s^G < e_s^* < e_s^P$ and $i_s^G \leq i_s^P < i_s^*$ for every $\lambda \in (0, 1]$, with $i_s^G = i_s^P$ only at $\lambda = 1$. The private manager cuts costs past the efficient level because its condition omits the marginal harm, and it makes fewer quality improvements than is efficient because it keeps half of each. The public conditions scale both returns by $\lambda/2 \leq 1/2$, so the public manager does less of both. The public efforts rise with λ and go to zero as $\lambda \to 0$, while the private efforts do not depend on λ.

Mortality. Let $a_s \geq 0$ measure how much service s matters for survival, and write mortality under regime $r \in \{G, P\}$ as

$$m^r = \bar{m} - \sum_s a_s \left[v_s(i_s^r) - b_s(e_s^r)\right] + u, \tag{C.4}$$

where $\bar{m} + u$ is mortality before either innovation, with u collecting the determinants of mortality outside healthcare, which do not depend on the control regime. In the data u varies across counties and years, and the identifying assumptions of Section 4 require its average change to be the same in counties that entered as in counties that had not yet entered.

Proposition 1 *Let $a_s > 0$ for some service, and let*

$$\Delta m \equiv m^P - m^G = \sum_s a_s \Big\{ \left[b_s(e_s^P) - b_s(e_s^G)\right] - \left[v_s(i_s^P) - v_s(i_s^G)\right] \Big\}. \tag{C.5}$$

(a) If $\lambda = 1$, mortality is higher under private control: $\Delta m > 0$. (b) $\Delta m > 0$ on an interval

$(\underline{\lambda}, 1]$ *for some* $\underline{\lambda} \in [0, 1)$. *(c) As* $\lambda \to 0$, $\Delta m \to \sum_s a_s \big[b_s(e_s^P) - v_s(i_s^P)\big]$. *If the survival value of the private manager's quality improvement exceeds the survival harm of its cost-cutting, so that this limit is negative, mortality is lower under private control for* λ *close enough to* 0*, and* Δm *changes sign at least once in* $(0, 1)$.

Proof of Proposition 1. Differencing (C.4) across regimes gives (C.5). (a) At $\lambda = 1$ the two quality conditions in (C.3) coincide, so $i_s^G = i_s^P$. The cost conditions give $c_s'(e_s^P) = 1 < 2 + b_s'(e_s^G) = c_s'(e_s^G)$, so $e_s^G < e_s^P$ because c_s' is decreasing, and $b_s(e_s^P) > b_s(e_s^G)$; with $a_s > 0$ for some s, $\Delta m > 0$. (b) Each public effort is continuous in λ, because its first-order condition in (C.3) is continuous in λ and strictly decreasing in effort, so Δm is continuous too, and being positive at $\lambda = 1$ it stays positive on an interval below 1. (c) The public conditions require $c_s'(e_s^G) - b_s'(e_s^G) = \beta_s'(i_s^G) = 2/\lambda$, and since both marginal returns are decreasing and finite at every positive effort, both public efforts go to zero as $\lambda \to 0$; $b_s(0) = v_s(0) = 0$ gives the limit. A negative limit and part (a) give the sign change by continuity. ■

At any λ, the sign of Δm compares the survival-weighted harm of the additional cost-cutting with the survival-weighted value of the additional quality improvement, so patients lose under private control when the harm is larger and gain when the value is. The parameter λ, the weights a_s, and the functions v_s are not observed, so these comparisons cannot be tested, and the estimates indicate only which case of Proposition 1 the evidence fits.

Inputs. To connect effort to employment and capacity, let the quantity of input j, such as staff or facilities, used in service s be

$$x_{js} = \bar{x}_{js} - \ell_{js}(e_s) + \kappa_{js}(i_s), \qquad \ell_{js}' \geq 0. \tag{C.6}$$

Here $\ell_{js}(e_s)$ is the quantity that cost-reducing effort saves and $\kappa_{js}(i_s)$ the change that quality improvement brings, both zero without effort, so $\bar{x}_{js}$ is the quantity used without either innovation. The sign of κ_{js}' is free, so quality improvement can require the input or save

it. Because $e_s^P > e_s^G$ and $i_s^P \geq i_s^G$, private control lowers the input when the additional savings from cost cuts exceed the additional requirements of quality improvement, and it never raises the input when quality improvement also saves it. A shortfall in employment or establishments is therefore consistent with either kind of innovation and measures neither effort nor efficiency.

Mapping to the estimates. The shift of births from government to private hospitals (Section 5.1) shows that control of care moved. The prenatal-care estimates (Section 5.2) measure access, one of the dimensions of care that cost cuts can lower. Employment, establishments, and hospitals (Section 5.3) are the inputs x_{js}, whose change the model does not sign, and the increase in all-cause mortality fits the case $\Delta m > 0$ of Proposition 1. The four stages describe a single contracting period, so the model says nothing about how long effects take to appear.

D Data construction

The checks in this appendix assess data consistency and comparability, not the identifying assumptions.

The vital-statistics complement is transcribed from annual report scans. Each table is checked against printed area subtotals, island totals, and internal decompositions: government, private, and other institutional births against the place-of-birth total; and attended, unattended, and unspecified counts against institutional births. Uncertain cells are inspected in enlarged scans; missing cells are not inferred. Two exceptions remain. Because the 1990 natality table is absent from the image set, low-birth-weight and very-low-birth-weight counts use an earlier extraction, checked only to ensure each does not exceed births. Birth shares divide the reports' births by place of delivery by the Planning Board's births. The two sources agree closely but not in every county-year: in five, the government- and private-

hospital births exceed the Planning Board's births, so the shares there exceed one. These are Ceiba in 1991 (355 hospital births against 353 births) and in 1993 (384 against 364), Maricao in 1991 (96 against 19), Luquillo in 1993 (379 against 319), and San Sebastián in 1995 (695 against 666); Maricao's cell is why the 1991 shares in Table 1 add up to 104.5 percent. These counts are used as published.[21] Replication documentation records the protocol, annual checks, and patches. Agreement between overlapping publications establishes consistency, not independent registration coverage.

Merging the vital-statistics complement. Source flags trace each observation in the merged Planning Board and digitized reports to its publication. Only counts are merged; rates are calculated with the panel's own denominators. Digitized records supplement early coverage of the four causes incompletely reported by the Planning Board, including observations from 1990 to 1995.

Industry definitions. The three health care employment series in Appendix Table A2 cover different industries and workers. The QCEW series is NAICS 621, Ambulatory Health Care Services, which leaves out hospitals (NAICS 622) and nursing and residential care (NAICS 623), and it counts jobs covered by unemployment insurance at their workplace (U.S. Bureau of Labor Statistics, 2025). County Business Patterns reports SIC 80, the health services division as published, which covers offices of physicians, dentists and other practitioners (801–804), nursing and personal care (805), hospitals (806), laboratories (807), and allied services (809). Its employment refers to the pay period including March 12, and its components are published separately in too few counties to be netted out of the division, so its series is broader than the ambulatory ones (U.S. Census Bureau, 2000b). The Census series counts employed civilians aged 16 and over, the self-employed and workers outside unemployment-insurance coverage included, in the ambulatory categories of the health ser-

[21] In San Sebastián in 1995, the report's government- and private-hospital births, 436 and 259, sum to 695 against its own total of 665 hospital births, so one of the two counts is off by 30; both are used as transcribed.

vices group of the 1990 Census industrial classification: offices and clinics of physicians (812), dentists (820), chiropractors (821), optometrists (822), and health practitioners n.e.c. (830), and health services n.e.c. (840). It leaves out the group's hospitals (831) and nursing and personal care facilities (832), and its counts are weighted estimates from the Puerto Rico five-percent microdata samples of 1990 and 2000 (Ruggles et al., 2025).

E Institutional detail

The public delivery system before the reform. CDTs offered primary and emergency care, laboratory testing, radiology, and pharmacy services; by 1993, many also offered internal medicine, obstetrics and gynecology, and pediatrics. They housed vaccination, communicable-disease, and environmental-health programs, alongside federally funded maternal and child health and agricultural migrant programs. Family Health Centers added dental care, social work, nutrition, and epidemiological nursing, with ambulatory teams comprising a physician, public-health nurse, and health assistant. Sub-regional and regional hospitals also housed many accredited residency programs (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). Patients entered through the public health center of their county and were referred from there to an area or regional hospital and, when necessary, to one of three medical centers (Pagán-Berlucchi and Muse, 1983).

Medicaid before the reform. Medicaid reimbursed the Department of Health an all-inclusive rate per inpatient day or outpatient visit, with no fee-for-service payment to physicians, and its recipients carried a Medicaid card. Public funds paid for Medicaid's categorically and medically needy recipients and for a third group of indigent residents who were ineligible for Medicaid and lacked health insurance, whose care a 1969 Commonwealth law guaranteed with Commonwealth funds alone. In 1979, about 1.3 million of the island's roughly 3.2 million residents were eligible for and enrolled in Medicaid. That year the island had 30,462 practicing health professionals, 5,368 of them physicians, and most physicians practiced privately. Public physicians earned about $15,000 to $30,000 a year in early-1980s dollars, against an estimated $45,000 to $50,000 average in private practice (Pagán-Berlucchi and Muse, 1983).

Who relied on the public system in 1990. In 1990, about 1.2 million residents had private insurance, 0.3 million had Medicare Parts A and B, and 0.2 million were public employees whose insurance premium the government partly paid. The other 1.8 million relied on public care, about 0.9 million of them through federal Medicaid, 0.3 million through the Commonwealth's own program and 0.6 million enrolled in neither. The Department of Health had about 1,000 physicians for these 1.8 million residents, while about 7,000 physicians served the other 1.7 million privately (Pan American Health Organization, 2007).

Where government and private hospitals were. In 1993, deaths occurred in government hospitals in 16 of the 78 counties and in private hospitals in 17, 22 counties in all, and the government and private hospitals of San Juan alone recorded 8.4 and 8.3 percent of all deaths on the island (Puerto Rico Department of Health, 2025); Appendix Figure A2, panel (b), maps these shares by county. In 1992, government hospitals accounted for 39,418 of the island's 64,137 hospital births, or 61.5 percent (Puerto Rico Department of Health, 2025).

Facility transfers and provider payment. The transfers began in 1994. Act 190 of September 5, 1996, did not affect the contracts signed before it and governed every later lease, sublease, sale, cession or other contract transferring the Department of Health's facilities to private interests. When the Secretary of Health chose the areas the government health plan would enter, the Department requested proposals for its facilities there. A provider taking over a facility had to commit to accept the coverage of the insurer the Health Insurance Administration contracted, and the Department's regular employees who did not join the new provider kept their rights under the Executive Reorganization Act of 1993 (Commonwealth of Puerto Rico, 1996). The CDTs were sold to private medical groups or nonprofit corporations, which became independent practice associations. By 2001, after many of the emergency rooms the CDTs had run closed, 58 of the 78 counties had one open around the clock (Pan

American Health Organization, 2007). Plans placed financial risk on primary-care physicians for the care other providers gave their enrollees, which Act 72 had not provided for, and the commission that evaluated the system in 2005 found that this risk faced primary-care physicians with the possibility of denying indicated services or delaying clinically necessary tests (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005; Pan American Health Organization, 2007).

The retained Medical Center. The Administration of Medical Services, which runs centralized clinical and support services for the Medical Center's hospitals, filled 2,068 positions across its three programs in December 1992 and 1,998 in September 1999 (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1999).

Plans and the order of entry. Each county's residents enrolled in the one plan serving their county, so enrollees had no choice of plan, and every plan covered the benefits the government defined (World Health Organization and Pan American Health Organization, 1998; Brugueras Fabre, 2002; Pan American Health Organization, 2007). The sources give the schedule of entry but not why the counties entered in that order. The 1998 cohort and San Juan had the largest populations and densities at baseline, while the 1994 cohort had the highest government share of births and more hospitals per resident than any cohort entering between 1995 and 1998 (Appendix Table A3). A county's cohort year is the first year in which the Health Insurance Administration counted enrollees in its region, so a county the reform reached late in a year is treated for that whole year.

Eligibility, take-up and coverage after the reform. The commission that evaluated the health system in 2005 describes the 200 percent income limit as a rise from 100 percent of the poverty line (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). Among eligible residents, take-up was 86 percent in 1995 and 1996, 93–98 percent from 1997 through 2005, and 90 percent in 2006 and 2007 (Administración de

Seguros de Salud de Puerto Rico, 2026). Mental-health and pharmacy administration were subsequently contracted separately (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005).

Coverage before and after the reform. Table 2 compares the coverage of residents in 1990 and 2003. Enrollment in the government health plan peaked at 1,795,067 in December 2000 and was 1,521,848 in December 2003, as the number of eligible residents fell from 1,866,965 to 1,618,540 (Administración de Seguros de Salud de Puerto Rico, 2026). The 2003 count is about 39 percent of the 3.879 million residents that Pan American Health Organization (2007) reports for that year, close to the commission's 40 percent. Pan American Health Organization (2007) puts the 2003 share without any insurance at about 8 percent, and a separate household survey estimated 353,058 uninsured residents, or 9.10 percent of the non-institutionalized population (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005).

Training and the licensed workforce. Accredited residency programs in primary-care specialties fell from 32 to 23 over the decade, partly because ambulatory training sites were lost when hospitals and CDTs were transferred to private providers (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). Between the Planning Board counts of 1992 and 1995, licensed physicians fell from 8,443 to 6,764 and registered nurses from 16,848 to 12,541 (Appendix Table A1). I treat the three Planning Board waves as counted on one definition.

E.1 Cross-cohort spillovers

Enrollment in the government health plan required proof of residence, so residents of a county the reform had not reached could not enroll, but residents of a treated county could still use the public facilities of a county it had not yet reached. To measure how many

residents of treated counties had a daily route to those facilities, I combine the Census 2000 commuting flows of Puerto Rico's 906,973 resident workers from each county of residence to each county of work (U.S. Census Bureau, 2000a), which I hold fixed across years, with the plan's eligibility and enrollment counts each December (Administración de Seguros de Salud de Puerto Rico, 2026). To get the maximum number of residents who would seek public health care in a not-yet-treated county, I assume that coverage was independent of cross-cohort commuting and that every uncovered resident of a treated county who worked in a county the reform had not-yet-reached sought public health care there. Uncovered residents are those that did not receive Medicaid or lacked health insurance. Let T denote living in a treated county, U being uncovered, and Y working in a county the reform had not yet reached. By Bayes' theorem,

$$\Pr(Y \mid T, U) = \frac{\Pr(U \mid T, Y)\ \Pr(Y \mid T)}{\Pr(U \mid T)} = \Pr(Y \mid T),$$

because independence makes $\Pr(U \mid T, Y) = \Pr(U \mid T)$. An uncovered worker living in a treated county was therefore as likely to work in a not-yet-treated county as any worker there: 39.6 percent in 1994, 15.9 percent in 1998, and zero in 2000. The number of uncovered workers with that route is the number of workers commuting from treated to not-yet-treated counties times $\Pr(U \mid T)$. I measure $\Pr(U \mid T)$ as the eligible residents who had not enrolled, as a share of the treated counties' 1990 Census population (U.S. Census Bureau, 1999), plus 4.5 percent for residents who were neither eligible nor insured. That 4.5 percent is their share in 2003, the 7 percent of residents without a plan that year (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005) less the 2.5 percent who were eligible but not enrolled (Administración de Seguros de Salud de Puerto Rico, 2026; Pan American Health Organization, 2007). In 1994, about 68,000 workers commuted from the 18 treated counties to the other 60, and 7.8 percent of the treated counties' residents were uncovered: 3.3 percent eligible but not enrolled, with 1994 eligibility imputed from the

1995 take-up rate of 85.9 percent, plus 4.5 percent. About 5,300 uncovered workers therefore had a daily route to public care in a not-yet-treated county. In 1998, about 124,000 workers commuted from the 77 treated counties to San Juan, and 6.9 percent of residents were uncovered (2.4 plus 4.5 percent), so about 8,600 did. The number peaked at about 15,800 in 1996, when 14 percent of the eligible had not yet enrolled, and it was zero in 2000. As a share of all workers living in treated counties, uncovered workers with this route never exceeded 4 percent. These are upper bounds, because not every such worker would have sought care away from home.